

CHAPTER 11

ESTIMATION

11.1 INTRODUCTION

Estimation is the process of extracting information about the value of a certain population parameter θ from the data. Consecutive observations (data points) $x_1, x_2, \dots$ are selected at random from the population being studied and are considered values of some random variables $X_1, X_2, \dots$. So if the same experiment of taking n observations were to be repeated, the new data points would most likely be different. The statistical laws that govern the variability of data under repetition (actual or hypothetical) can be used to build the theory of statistical inference. In estimation theory, it is usually assumed that the distribution of each observation X_i is known, except for the value of some parameter θ . An estimator of θ is then a rule that allows us to calculate an approximation of θ , based on sample $X_1, \dots, X_n$.

■ EXAMPLE 11.1

Let $X_1, X_2, \dots$ represent the weights of successive trout caught at a certain location. We are interested in a parameter such as $\theta = P\{X_i > w_0\}$, where w_0 is some fixed weight, so that θ is the fraction of trout whose weight exceeds a given threshold w_0 .

Given the observed weights $X_1, \dots, X_n$, we want to estimate θ as

$$T_n = T(X_1, \dots, X_n) = \frac{\text{number of } i \text{ such that } X_i \geq w_0}{n}.$$

The formula above gives an obvious, and rather unsophisticated estimator of θ as a relative frequency.

There is another method of estimating the same parameter. Assume that the weights of trout follow a normal distribution $N(\mu, \sigma^2)$. Given the sample $X_1, \dots, X_n$, we then calculate the sample mean $\bar{X} = (1/n)(X_1 + \dots + X_n)$ and sample variance

$S^2 = [1/(n-1)] \sum (X_i - \bar{X})^2$. Both $\bar{X}$ and S^2 can be regarded as approximations of μ and σ^2 , respectively. Consequently, we estimate θ as

$$U_n = U(X_1, \dots, X_n) = 1 - \Phi\left(\frac{w_0 - \bar{X}}{S}\right), \quad (11.1)$$

where Φ is the cumulative distribution function (cdf) of a standard normal variable. Here the rationale is that

$$\theta = P\{X \geq w_0\} = P\left\{Z \geq \frac{w_0 - \mu}{\sigma}\right\} = 1 - \Phi\left(\frac{w_0 - \mu}{\sigma}\right),$$

which is approximated by (11.1).

The example above shows that there can be several estimators of the same parameter (i.e., several distinct rules of calculating an approximation of θ , given the sample). Since every estimator, such as T_n and U_n in Example 11.1, is a random variable, the obvious questions are:

1. How to assess the performance of estimators and to choose the best one?
2. Are there methods for obtaining estimators other than “ad hoc” methods, used to obtain estimators T_n and U_n in Example 11.1?

A systematic attempt to answer these two questions resulted in a theory with a clear conceptual structure, supported by powerful theorems. The empirical situation is such that we observe values of independent random variables $X_1, X_2, \dots$, sampled from the same distribution $f(x, \theta)$, where f is a density or a probability function, depending on whether the X_i 's are continuous or discrete. This distribution depends on a parameter that assumes some value θ (unknown to the observer) from a parameter space Θ . The examples abound: the X_i 's can be Bernoulli observations for an unknown probability of success θ , or normally distributed observations with an unknown mean θ and a known standard deviation, or with a known mean and an unknown standard deviation θ , and so on.

To answer question 1: How to assess estimators? Let $T_n = T_n(X_1, \dots, X_n)$ be the estimator for a random sample of size n , namely some function of observations $X_1, \dots, X_n$, selected to approximate θ . The first requirement for a “good” estimator is its consistency, defined by the requirement that $T_n \rightarrow \theta$. Since T_n 's are random variables, we must specify the type of convergence (e.g., in probability or almost surely). The need of consistency is obvious: estimators that are not consistent do not guarantee that one gets closer to the true value of the parameter by increasing the sample size.

If $E(T_n) = \theta$ for every n , then estimator T_n is called unbiased. This means that if the estimation were to be repeated several times for different samples (but with the use of the same estimator), the results would be, on average, “on target.” The “quality” of an unbiased estimator may then be defined as its variance. This corresponds to taking $(t_n - \theta)^2$ as the “penalty” or “loss” due to the error of accepting the value t_n of T_n as the approximation of θ . While the squared error is not the only loss function possible, it is realistic in many problems.

To sketch this development: First, one has a powerful Cramér–Rao inequality, which states that there exists a lower bound for variances of all unbiased estimators of θ (for any fixed sample size n). This bound shows therefore the best that can be achieved in estimating a given parameter, in a sense of providing a yardstick by which one can tell how close a given estimator is to the “ideal” (i.e., to the estimator with the smallest possible variance).

To answer question 2: How to construct estimators? There are methods (maximum likelihood, moments, etc.) of finding an estimator, and there are also methods of modifying

an estimator in order to improve it (e.g., to make its variance closer to the possible minimum value). One such improvement method is based on the concept of a sufficient statistic that retains information in the data that is relevant for estimating parameter θ . The Rao–Blackwell theorem says that if T is an unbiased estimator of θ , and S is a sufficient statistic for θ , then $T^* = E(T|S)$ is an estimator of θ that is better (or not worse) than T . Therefore, one gets a powerful tool of improving estimators: start with any unbiased estimator T , and find its conditional expectation T^* with respect to a sufficient statistic. If this new estimator T^* is not the best, then continue the process, conditioning with respect to another sufficient statistic, and so on. Instead of such conditioning “one step at a time,” one can condition T with respect to the so-called minimal sufficient statistic (i.e., maximal reduction of the data that preserves information about θ). The estimator T^* obtained by such conditioning cannot be improved any further. This raises the obvious questions: Is T^* the best possible estimator? Does the estimator T^* depend on the initial starting estimator T ? These two questions turn out to be closely related. If the family of distributions $\{f(x, \theta), \theta \in \Theta\}$ is complete, then T^* attains the lower bound given by the Cramér–Rao theorem, and also, T^* can be obtained by conditioning any unbiased estimator of θ with respect to the minimal sufficient statistic.

As already mentioned, the sketch above gives a “success story with a happy ending” in statistical methodology. Presented in this manner it may appear easy and effortless. But one should remember that bringing the theory to its present form required analysis of countless examples, proving or disproving various conjectures, formulating weaker and weaker sets of assumptions under which such or other assertion is true, and this took about half a century of effort on the part of many statisticians.

In this chapter, we will also present some other topics and results in estimation: asymptotic properties of maximum likelihood estimators, Bayesian estimation, the bootstrap approach, and so on.

Let us begin with a simple estimation problem and a number of possible solutions. This will allow us to formulate various questions, as well as to suggest some natural generalizations to be discussed later in this chapter.

In many situations where statistical methods apply, one observes a simple random sample $X_1, \dots, X_n$ from a distribution whose functional form is known, except for the value of a certain parameter θ . This means that the actual observations $x_1, \dots, x_n$ are the values of iid random variables $X_1, \dots, X_n$, each with a distribution that will be denoted $f(x, \theta)$, where f stands either for the density or for the probability function depending on whether X_i ’s are continuous or discrete.

We also assume that θ is an element of some *parameter space* Θ . In a simple scenario, θ is a single number, so Θ is a subset of the real line, but in general, Θ may be a multidimensional space, or even a space with a more complicated structure.

In what follows, we often assume that Θ is an interval of the real line, so that we can employ standard optimization techniques, for example, involving the differentiation of various quantities with respect to θ .

■ EXAMPLE 11.2

A politician needs to estimate the proportion of voters who favor a certain issue. In a public opinion poll, n persons are sampled from the population and their responses $X_1, \dots, X_n$ are noted, where $X_i = 1$ or 0 , depending whether or not the i th person polled favors the issue in question. Letting θ denote the proportion of voters in favor of the issue, we have

$$f(x, \theta) = P\{X_i = x|\theta\} = \begin{cases} \theta & \text{for } x = 1 \\ 1 - \theta & \text{for } x = 0. \end{cases}$$

Here $0 \leq \theta \leq 1$ (i.e., $\Theta = [0, 1]$), and $f(x, \theta)$ can be written as:

$$f(x, \theta) = \theta^x (1 - \theta)^{1-x}, \quad x = 0, 1.$$

■ EXAMPLE 11.3

Measurement with error is often represented as an estimation problem. We are to measure the value of an attribute of an object (its weight, dimension, temperature, content of some substance, etc.). The true value of this attribute is θ , but the measurements are subject to error. For instance, it is often assumed that the i th measurement is $X_i = \theta + \epsilon_i$, where $\epsilon_1, \epsilon_2, \dots$ are iid random variables. If moreover $\epsilon_i \sim N(0, \sigma^2)$, then $X_i \sim N(\theta, \sigma^2)$, and

$$f(x, \theta) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\theta)^2/2\sigma^2},$$

where σ is the standard deviation of measurements (assumed known). In this case, the parameter space Θ is the subset of a real line representing the possible values of the measured attribute.

If the standard deviation of measurement, σ , is unknown, then θ is a two-dimensional parameter $\theta = (\mu, \sigma)$, and we may be interested in estimating one or both components of θ . In this case, the parameter space Θ is a subset of the half-plane $\{(\mu, \sigma) : \sigma > 0\}$.

Before proceeding further, a few comments about notation and terminology are in order. First, we will typically use a symbol such as T , to denote a statistic, even if it refers to various possible sample sizes. Consider, for instance, a sample mean $\bar{X} = (1/n)(X_1 + \dots + X_n)$. In fact, however, we are considering here a *sequence* of statistics, each being a function of a different number of arguments. To formulate this properly, it will sometimes be convenient to use the notation involving the sample size. We will write T_n for the estimator based on a sample of size n so that $T_n = T_n(X_1, \dots, X_n)$.

Second, we will call a statistic an *estimator* (of θ), when this statistic is *used to estimate* θ . Thus, formally, when sample size n is not specified, an estimator is a sequence of statistics, the n th one depending on observations $X_1, \dots, X_n$. The value of an estimator, obtained for a particular sample, will be called an *estimate* of θ .

Finally, in presenting the general theory, we will use the symbol θ for the unknown parameter. However, we will also be using traditional notation, such as p for probability of success and σ for standard deviation.

The following example will show several estimators of the same parameter:

■ EXAMPLE 11.4

Suppose that we take a random sample $X_1, \dots, X_n$ from the $U[0, \theta]$ distribution so that

$$f(x, \theta) = \begin{cases} 1/\theta & \text{if } 0 \leq x \leq \theta \\ 0 & \text{otherwise.} \end{cases}$$

The objective is to estimate the range θ .

SOLUTION. We will suggest several estimators of θ . First, we get some information about θ from the largest element of the sample. We feel (and will justify it later) that

as the sample size increases, the largest value should get closer and closer to θ . This suggests taking

$$T_1 = X_{n:n} \quad (11.2)$$

as an estimator of θ . A disadvantage of T_1 is that it always underestimates θ . We can remedy this in a number of ways. For instance, we can argue as follows: In the sample of size n , observed values $X_1, \dots, X_n$ partition the interval $[0, \theta]$ into $n + 1$ intervals. Since $X_1, \dots, X_n$ tend to be “evenly dispersed” over $[0, \theta]$, each of these $n + 1$ intervals will have, on average, the same length $\theta/(n + 1)$. Thus, we should “push” T_1 to the right by $(1/n)T_1$, which suggests using the estimator

$$T_2 = \left(1 + \frac{1}{n}\right) T_1 = \frac{n+1}{n} X_{n:n}. \quad (11.3)$$

Another way of “adjusting” T_1 may be based on the fact that by symmetry, the maximal observation is, on average, at the same distance from the upper bound θ as the minimal observation is from the lower bound 0. Thus, we can use the estimator

$$T_3 = X_{1:n} + X_{n:n}. \quad (11.4)$$

We will also argue that the minimum of the sample allows us to estimate θ . The same argument as above, with a partition of the range $[0, \theta]$ into $n + 1$ parts of about equal length, suggests the estimator

$$T_4 = (n + 1)X_{1:n}. \quad (11.5)$$

This estimator does not seem very reliable. In particular, it may happen that $T_4 < X_{n:n}$, in which case using T_4 would make no sense. Moreover, maximum and minimum of the sample have—by symmetry—the same variance. The estimator T_4 magnifies this variance by the factor $(n + 1)^2$. We will show later how this fact affects the precision of the estimators.

Finally, we take

$$T_5 = 2\bar{X}, \quad (11.6)$$

arguing that the average $\bar{X}$ should be close to the midpoint $\theta/2$.

Example 11.4 shows that one can have several estimators for the same parameter. This poses a natural question of establishing criteria for a choice. The next few sections give criteria to evaluate the performance of estimators.

Another question concerns the methods of finding estimators, especially “good” ones: Rather than rely on intuition and common sense, it is desirable to have a scheme that can produce estimators in an “automatic” fashion. We will also present some modern generalizations and extensions of estimation procedures, especially computer-intensive techniques called *resampling methods*.

11.2 CONSISTENCY

One of the basic properties of a good estimator is that it provides more precise information about θ with the increase of the sample size n . We introduce the following definition:

Definition 11.2.1 The estimator $T = \{T_n, n = 1, 2, \dots\}$ of parameter θ is called *consistent*, if T_n converges to θ in probability, that is,

$$\lim_{n \rightarrow \infty} P\{|T_n - \theta| \leq \epsilon\} = 1 \quad (11.7)$$

for every $\epsilon > 0$. The estimator T_n will be called *strongly consistent* if T_n converges to θ almost surely,

$$P \left\{ \lim_{n \rightarrow \infty} T_n = \theta \right\} = 1. \quad (11.8)$$

When both kinds of consistency are considered at the same time, estimators satisfying (11.7) are called *weakly consistent*. $\square$

We will now analyze the consistency of the five estimators $T_1 - T_5$ introduced in Example 11.4.

■ EXAMPLE 11.5

The distribution of $T_1 = X_{n:n}$ is easy to obtain. Indeed, since X_i 's are uniform on $[0, \theta]$, we have

$$P\{X_i \leq t\} = F(t) = \begin{cases} 0 & \text{if } t < 0 \\ t/\theta & \text{if } 0 \leq t \leq \theta \\ 1 & \text{if } t > \theta. \end{cases}$$

We are interested only in probabilities for t between 0 and θ . Since $X_{n:n} \leq t$ if and only if $X_i \leq t$ for all i , by independence, we have

$$P\{T_1 \leq t\} = P\{X_1 \leq t, \dots, X_n \leq t\} = \left(\frac{t}{\theta}\right)^n. \quad (11.9)$$

For $0 < \epsilon < \theta$,

$$P\{|T_1 - \theta| \leq \epsilon\} = P\{T_1 \geq \theta - \epsilon\} = 1 - \left(\frac{\theta - \epsilon}{\theta}\right)^n.$$

Since $[(\theta - \epsilon)/\theta]^n \rightarrow 0$ as $n \rightarrow \infty$, estimator T_1 is consistent. Using notation from Chapter 9, we may write

$$T_1 \xrightarrow{P} \theta.$$

■ EXAMPLE 11.6

Next, $T_2 = [(n+1)/n] T_1$, and since $(n+1)/n \rightarrow 1$, we have $T_2 \xrightarrow{P} \theta$.

■ EXAMPLE 11.7

Regarding the consistency of estimator T_3 , observe first that for $0 < \epsilon < \theta$,

$$P\{X_{1:n} > \epsilon\} = \prod_{i=1}^n P\{X_i > \epsilon\} = \prod_{i=1}^n [1 - F(\epsilon)] = \left(1 - \frac{\epsilon}{\theta}\right)^n \rightarrow 0, \quad (11.10)$$

which means that $X_{1:n} \xrightarrow{P} 0$. Consequently, since

$$T_3 = T_1 + X_{1:n} \xrightarrow{P} \theta,$$

T_3 is also consistent.

■ EXAMPLE 11.8

Estimator T_4 seems inferior to the others, as remarked in Example 11.4. We will show that T_4 is not consistent. For $0 < \epsilon < \theta$, we have

$$\begin{aligned} P\{|T_4 - \theta| \leq \epsilon\} &= P\{|(n+1)X_{1:n} - \theta| \leq \epsilon\} = P\left\{\frac{\theta - \epsilon}{n+1} \leq X_{1:n} \leq \frac{\theta + \epsilon}{n+1}\right\} \\ &= P\left(X_{1:n} \leq \frac{\theta + \epsilon}{n+1}\right) - P\left(X_{1:n} \leq \frac{\theta - \epsilon}{n+1}\right). \end{aligned}$$

The cdf of $X_{1:n}$ can be obtained from formula (11.10). Using the fact that $(1 + x/n)^n \rightarrow e^x$, we then have

$$\begin{aligned} P\{|T_4 - \theta| \leq \epsilon\} &= 1 - \left(1 - \frac{\theta + \epsilon}{\theta(n+1)}\right)^n - \left[1 - \left(1 - \frac{\theta - \epsilon}{\theta(n+1)}\right)^n\right] \\ &= \left(1 - \frac{\theta - \epsilon}{\theta(n+1)}\right)^n - \left(1 - \frac{\theta + \epsilon}{\theta(n+1)}\right)^n \\ &\rightarrow e^{-(\theta - \epsilon)/\theta} - e^{-(\theta + \epsilon)/\theta} < 1. \end{aligned}$$

Thus, $\lim_{n \rightarrow \infty} P\{|T_4 - \theta| > \epsilon\} > 0$, which shows that T_4 is not a consistent estimator of θ .

■ EXAMPLE 11.9

By the law of large numbers, we have

$$\bar{X} \xrightarrow{P} E(X_i) = \frac{\theta}{2}, \quad (11.11)$$

so

$$T_5 = 2 \bar{X} \xrightarrow{P} 2 \times \frac{\theta}{2} = \theta, \quad (11.12)$$

which shows that the estimator T_5 is also consistent.

PROBLEMS

11.2.1 Let $X_1, \dots, X_n$ be a random sample from BIN(1, θ) distribution and let $S_n = (1/n) \sum X_i$. Show consistency of estimators: (i) $\hat{\theta} = S_n/n$. (ii) $\widehat{\text{Var}}(X) = (S_n/n)(1 - S_n/n)$.

11.2.2 Let $X_1, \dots, X_n$ be a random sample from the distribution with density $f(x; \theta) = e^{-(x-\theta)}$ for $x \geq \theta$, and $f(x, \theta) = 0$ otherwise. Check if $T = X_{1:n}$ is a consistent estimator of θ .

11.2.3 The density of a Pareto distribution is $f(x; \alpha, \theta) = \alpha \theta^\alpha x^{-(\alpha+1)}$ for $x \geq \theta, \theta > 0$, and $\alpha > 0$. Show that $T = X_{1:n}$ is a consistent estimator of θ .

11.2.4 Assume that the observations are taken from the $U[0, \theta]$ distribution, and let U_n be the number of observations (out of first n) that are less than 5. Show that if $\theta > 5$, then $T_n = 5n/U_n$ is a consistent estimator of θ .

11.2.5 Observations are randomly sampled from the $U[0, \theta]$ distribution. After the sample size reaches n , the experimenter starts recording the minimum observations $X_{1:n}, X_{1:n+1}, \dots$. He will continue until he gets X_{n+N} such that

$$X_{1:n} = X_{1:n+1} = \dots = X_{1:n+N-1} > X_{1:n+N}.$$

Suggest an estimator of θ based on observing $X_{1:n}, \dots, X_{1:n+N}$. [Hint: Find the probability distribution of N and express $E(N)$ as a function of θ .]

11.3 LOSS, RISK, AND ADMISSIBILITY

Assume again that n independent observations $X_1, \dots, X_n$ of a random variable with distribution²¹ $f(x, \theta)$ are taken in order to estimate θ . The closer θ^* and θ will be, the more successfully will θ^* subsequently serve some specific purpose.

In the simplest case, this degree of success may be a decreasing function of the error $|\theta^* - \theta|$ so that the smaller the error, the better. One can easily imagine, however, situations where the error of overestimating θ (i.e., $\theta^* > \theta$) is less serious than the error of underestimating it ($\theta^* < \theta$). To build a general theory, we assume that such a situation can be adequately represented by specifying the loss function $\mathcal{L}(\theta^*, \theta)$, which describes the negative consequences of proceeding as if the value of the parameter were θ^* while in reality it is θ (we consider “loss” and negative consequences, but by changing the sign, we can convert the considerations to “rewards” and positive consequences).

Suppose now that the experimenter decides to use the estimator $T = T(X_1, \dots, X_n)$ with the loss equal to $\mathcal{L}(T, \theta)$. The performance of estimator T can be evaluated as the average loss

$$R_T(\theta) = E_\theta\{\mathcal{L}(T(X_1, \dots, X_n), \theta)\}, \quad (11.13)$$

where E_θ stands for expected value with respect to the distribution $f_n(x, \theta)$. The function R given by formula (11.13) above is called a *risk function*.

We now need to introduce two important concepts:

Definition 11.3.1 The estimator T_1 is *R-dominating* estimator T_2 , or is *R-better* than T_2 , if for all $\theta \in \Theta$ we have

$$R_{T_1}(\theta) \leq R_{T_2}(\theta),$$

and the inequality is strict for at least one value of θ .

Moreover, an estimator T is called *R-inadmissible* if there exists an estimator T' that is *R-better* than T . Otherwise, T will be called *R-admissible*. $\square$

The basis for evaluation of an estimator is the risk function R , which depends on the unknown parameter θ . Clearly, if we have two estimators, and their risk functions are such that one of them is below the other (or equal to it) *regardless of the value of θ* , we can decide that the corresponding estimator is better. Quite often, however, the two risk functions cross each other (i.e., one is below the other for some θ , and above it for some other θ). In such a case, the estimators are *not comparable*: Because we do not know θ , and we do not know which risk function is smaller at the actual (true) value of θ , we cannot decide which estimator is better. Thus, we obtain only a *partial order* of estimators; estimators that are not dominated by any other are admissible. Within the class of admissible estimators, by definition, none dominates the others, and therefore we still need other criteria for choice. But

²¹ We remind the reader that $f(x, \theta)$ is either the density or the probability function.

at least the problem becomes reduced in the sense that all inadmissible estimators are ruled out. In practice, the search for the class of admissible estimators for a specific loss function can be difficult.

To build a general theory, it is necessary to choose a loss function that is acceptable. In 1821, Gauss proposed a *squared error*

$$\mathcal{L}(\theta^*, \theta) = (\theta^* - \theta)^2 \quad (11.14)$$

as a measure of loss for its mathematical simplicity and convenience. However, the use of this loss function makes sense only for estimators with finite second moment (i.e., finite variance). We shall tacitly make this assumption. The risk of an estimator T for loss (11.14) is therefore $E_\theta[T(X_1, \dots, X_n) - \theta]^2$. This risk appears so frequently that it has acquired its own name.

Definition 11.3.2 The risk of an estimator T computed for the loss function (11.14) is called the *mean squared error* (MSE) of T , and is typically denoted as

$$\text{MSE}_\theta(T) = E_\theta[T(X_1, \dots, X_n) - \theta]^2. \quad (11.15)$$

Also, when no risk function is specified, admissibility will always mean admissibility with respect to the MSE. $\square$

Another often used loss function, $\mathcal{L}(\theta^*, \theta) = |\theta^* - \theta|$, was suggested by Laplace at the beginning of the nineteenth century. It leads to a theory that is less tractable mathematically, and has only begun to be developed recently with powerful computers becoming widely available. While both loss functions, $(\theta^* - \theta)^2$ and $|\theta^* - \theta|$, are symmetric, there are situations in which overestimating the parameter has different consequences than underestimating it. Think about coming to the airport ahead of time and having to wait or coming too late and missing your flight, paying your tuition in advance instead of getting a bill for late fees since you did not pay by the deadline.

Several different asymmetric loss functions have been proposed in the literature to deal with such cases, for example,

$$\mathcal{L}(\theta^*, \theta) = \begin{cases} a(\theta^* - \theta)^2 & \text{if } \theta^* \geq \theta \\ (\theta^* - \theta)^2 & \text{if } \theta^* < \theta \end{cases}$$

for some constant a , or a linear exponential (*linex*) loss function

$$\mathcal{L}(\theta^*, \theta) = \exp\{a(\theta^* - \theta)\} - a(\theta^* - \theta) - 1,$$

where $a \neq 0$, or a *zero-one* loss function

$$\mathcal{L}(\theta^*, \theta) = \begin{cases} 0 & \text{if } \theta^* = \theta \\ 1 & \text{if } \theta^* \neq \theta. \end{cases}$$

Asymmetric loss functions are often used in various optimization procedures, for example in decision theory, forecasting, classification, or machine learning²².

²²One could argue, however, that an adequate loss function should reflect consequences of errors made *after* the parameter has been estimated (see Gafrikova and Niewiadomska-Bugaj, 1992). The starting point in the latter approach is that we use the distribution $f(x, \theta)$ for some purpose, such as prediction, calculation of some probabilities, and so on. Thus, the loss $\mathcal{L}(\theta^*, \theta)$ should depend on how much the (estimated) distribution $f(x, \theta^*)$ differs from the true distribution $f(x, \theta)$. Such “difference” between distributions can be expressed through a measure of difficulty in discriminating between the two distributions.

The following definition and theorem show the important role of the first two moments of T in the theory based on MSE.

Definition 11.3.3 An estimator T such that

$$E_\theta(T) = \theta \quad (11.16)$$

for every θ will be called *unbiased*, and the difference

$$B_\theta(T) = E_\theta(T) - \theta \quad (11.17)$$

will be called the *bias* of estimator T . $\square$

Clearly, $B_\theta(T) = 0$ if and only if T is unbiased. The estimator T will be called *positively* (or *negatively*) *biased*, depending on whether $B_\theta(T) > 0$ or $B_\theta(T) < 0$. More generally, if

$$\lim_{n \rightarrow \infty} E_\theta[T(X_1, \dots, X_n)] = \theta, \quad (11.18)$$

then T will be called *asymptotically unbiased*.

We have the following theorem:

Theorem 11.3.1 *The MSE of an estimator is the sum of its variance and square of the bias.*

Proof:

$$\begin{aligned} \text{MSE}_\theta(T) &= E_\theta(T - \theta)^2 = E_\theta\{[T - E_\theta(T)] + [E_\theta(T) - \theta]\}^2 \\ &= E_\theta[T - E_\theta(T)]^2 + [E_\theta(T) - \theta]^2 \\ &\quad + 2E_\theta\{[T - E_\theta(T)] \times [E_\theta(T) - \theta]\} \\ &= \text{Var}_\theta(T) + [B_\theta(T)]^2. \end{aligned} \quad \square$$

Consequently, the MSE of an unbiased estimator is equal to its variance.

We will now find bias and MSE for the estimators $T_1 - T_5$ introduced in Example 11.4.

Observe first that for $n = 1$, we have $T_1 = X_1, T_2 = T_3 = T_4 = T_5 = 2X_1$. Next $E_\theta(T_1) = \theta/2$, whereas the remaining estimators are unbiased. Finally, $\text{Var}(T_1) = \text{Var}(X_1) = \theta^2/12$; hence the MSE of all of those estimators are the same, $\theta^2/3$. They are either all admissible or all inadmissible if there exists a better estimator. In the examples below, we assume that $n > 1$.

■ EXAMPLE 11.10

Let us begin with the estimator $T_1 = X_{n:n}$. Clearly, since $T_1 < \theta$, the estimator is biased. The cdf of T_1 is given by (11.9), so the density of T_1 is

$$f_{T_1}(t) = \frac{nt^{n-1}}{\theta^n}, \quad 0 \leq t \leq \theta, \quad (11.19)$$

and $f_{T_1}(t) = 0$ otherwise. Thus,

$$E_\theta(T_1) = \frac{n}{\theta^n} \int_0^\theta t^n dt = \frac{n}{n+1} \theta, \quad (11.20)$$

a result that we derived in Example 9.3 and used to justify the need to “adjust” T_1 to obtain unbiased estimators T_2 and T_3 .

The bias of T_1 is negative:

$$B_\theta(T_1) = \frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}. \quad (11.21)$$

Using the formula for $\text{Var}(T_1)$ obtained in Example 9.3, we obtain

$$\text{MSE}_\theta(T_1) = \text{Var}_\theta(T_1) + [B_\theta(T_1)]^2 = \frac{2\theta^2}{(n+1)(n+2)}. \quad (11.22)$$

■ EXAMPLE 11.11

Estimator $T_2 = [(n+1)/n]T_1$ is unbiased. Therefore,

$$\text{MSE}_\theta(T_2) = \text{Var}_\theta(T_2) = \frac{(n+1)^2}{n^2} \text{Var}_\theta(T_1) = \frac{\theta^2}{n(n+2)}. \quad (11.23)$$

Comparing with $\text{MSE}_\theta(T_1)$, we have

$$\frac{\text{MSE}_\theta(T_2)}{\text{MSE}_\theta(T_1)} = \frac{n+1}{2n} < 1,$$

which means that the risk of T_2 is, for large n , about half of the risk of T_1 . This result shows that the estimator T_1 is not admissible: There exists an estimator that is better for all θ , namely T_2 (this does *not* imply that T_2 is admissible!).

■ EXAMPLE 11.12

Let us now consider the estimator $T_3 = X_{1:n} + X_{n:n}$. This modification of T_1 is again intended to remove the bias so that $B_\theta(T_3) = 0$. Indeed, $E(X_{n:n}) = [n/(n+1)]\theta$, and by symmetry, $E(X_{1:n}) = [1/(n+1)]\theta$. It was shown in Section 9.3 that

$$\text{Var}_\theta(X_{1:n}) = \text{Var}_\theta(X_{n:n}) = \text{Var}_\theta(T_1).$$

Based on formula (9.35), we obtain

$$\begin{aligned} \text{MSE}_\theta(T_3) &= \text{Var}_\theta(T_3) \\ &= \text{Var}_\theta(X_{1:n}) + \text{Var}_\theta(X_{n:n}) + 2\text{Cov}_\theta(X_{1:n}, X_{n:n}) \\ &= 2\text{Var}_\theta(T_1) + 2\text{Cov}_\theta(X_{1:n}, X_{n:n}) \\ &= \frac{2n\theta^2}{(n+1)^2(n+2)} + \frac{2\theta^2}{(n+1)^2(n+2)} \\ &= \frac{2\theta^2}{(n+1)(n+2)}. \end{aligned} \quad (11.24)$$

Thus, $\text{MSE}_\theta(T_3) > \text{MSE}_\theta(T_2)$; again, T_3 is not admissible.

■ EXAMPLE 11.13

Estimator $T_4 = (n+1)X_{1:n}$ is unbiased so that

$$\text{MSE}_\theta(T_4) = \text{Var}_\theta(T_4) = (n+1)^2 \text{Var}_\theta(X_{1:n}) = \frac{n\theta^2}{n+2}.$$

Compared with T_2 , we see that T_4 is inadmissible—observe that the variance of T_4 does not even tend to 0.

■ EXAMPLE 11.14

For the estimator T_5 , since $\text{Var}(X_i) = \theta^2/12$, we have

$$\text{Var}(T_5) = \text{Var} \left[\frac{2}{n}(X_1 + \cdots + X_n) \right] = \frac{4}{n^2} \times n \text{Var}(X_i) = \frac{\theta^2}{3n}.$$

This time, $\text{MSE}_\theta(T_2)/\text{MSE}_\theta(T_5) = 3/(n+2)$, which for $n > 1$ is less than 1, and therefore T_5 is not admissible.

It follows from the examples above that for $n > 1$, the best out of the five suggested estimators is T_2 : In each case, the risk is a parabola $k\theta^2$, with the coefficient k smallest for estimator T_2 . As we concluded, T_1, T_3, T_4 , and T_5 are inadmissible. This, however, does not imply automatically that T_2 is admissible. There may exist an estimator with a smaller risk than the risk of T_2 . To determine the admissibility of T_2 , let us find the coefficient α such that $T = \alpha X_{n:n} = \alpha T_1$ has the MSE smaller than T_2 . We have

$$E_\theta(T) = \alpha E(T_1) = \frac{\alpha n}{n+1} \theta,$$

and therefore

$$B_\theta(T) = \frac{\alpha n}{n+1} \theta - \theta = \frac{\alpha n - (n+1)}{n+1} \theta.$$

Since

$$\text{Var}_\theta(T) = \alpha^2 \text{Var}_\theta(T_1) = \frac{\alpha^2 n}{(n+1)^2(n+2)} \theta^2,$$

we have

$$\begin{aligned} \text{MSE}_\theta(T) &= \left[\frac{\alpha^2 n}{(n+1)^2(n+2)} + \frac{\alpha^2 n^2 - 2\alpha n(n+1) + (n+1)^2}{(n+1)^2} \right] \theta^2 \\ &= \left[\frac{\alpha^2 n + (n+2)(\alpha^2 n^2 - 2\alpha n(n+1) + (n+1)^2)}{(n+1)^2(n+2)} \right] \theta^2. \end{aligned}$$

Differentiating the numerator with respect to α , we obtain $\alpha = (n+2)/(n+1)$ as the value that minimizes $\text{MSE}_\theta(T)$. For such α , the estimator

$$T = \frac{n+2}{n+1} X_{n:n} \tag{11.25}$$

has the risk

$$\text{MSE}_\theta(T) = \frac{\theta^2}{(n+1)^2} < \frac{\theta^2}{n(n+2)} = \text{MSE}_\theta(T_2).$$

Thus, T_2 is also inadmissible, and the best estimator (of those considered) is the estimator T given by (11.25).

PROBLEMS

- 11.3.1** Let T_1 and T_2 be two unbiased estimators of θ with variances σ_1^2, σ_2^2 , respectively. Find values a and b such that: **(i)** Estimator $aT_1 + bT_2$ is unbiased. **(ii)** Unbiased estimator $aT_1 + bT_2$ has a minimum variance assuming that T_1 and T_2 are independent. **(iii)** Unbiased estimator $aT_1 + bT_2$ has a minimum variance if $\text{Cov}(T_1, T_2) = C$.

- 11.3.2** Let $X_1, \dots, X_n$ be a random sample from a $N(\theta, \sigma^2)$ distribution with σ^2 known. Show that the estimator T of θ , defined as $T(X_1, \dots, X_n) \equiv 3$ ($T = 3$ regardless of the observations), is admissible.
- 11.3.3** Let $X_1, \dots, X_n$ be a random sample from $\text{EXP}(1/\theta)$ distribution. Obtain the mean squared errors of two estimators of θ : $T_1 = \bar{X}$ and $T_2 = [n/(n+1)]\bar{X}$.
- 11.3.4** Let $X_1, \dots, X_n$ be n Bernoulli trials with probability of success θ , and let $S = \sum_{i=1}^n X_i$. Compare the mean squared errors of two estimators of θ : $T_1 = S/n$ and $T_2 = (S+1)/(n+2)$.
- 11.3.5** Let $Y_1, \dots, Y_n$ be a random sample of size n from the discrete distribution with probability function $f(y; \theta) = \theta(1-\theta)^{y-1}$, $y = 1, \dots$. Compare the MSE's of two estimators of $\tau = 1/\theta$: $T_1 = \bar{Y}$ and $T_2 = [n/(n+1)]\bar{Y}$.
- 11.3.6** Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, \sigma^2)$ distribution (μ and σ^2 are unknown), and let

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2, \quad S_1^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

be two estimators of σ^2 . **(i)** Compare the MSE's of S^2 and S_1^2 . **(ii)** Consider

$$S_k^2 = k \sum_{i=1}^n (X_i - \bar{X})^2$$

as estimators of σ^2 and find k for which S_k^2 has smallest MSE. Explain why, in practice, the only values of k used are (suboptimal): $k = 1/n$ and $k = 1/(n-1)$.

- 11.3.7** Let $U = X_{1:n}$ and let $V = X_{n:n}$ in a random sample from $U[\theta-1, \theta+1]$ distribution. **(i)** Show that $\bar{X}$ and $(U+V)/2$ are both unbiased estimators of θ . **(ii)** Determine the MSE's of estimators in (i).
- 11.3.8** Let $X_1, \dots, X_4$ be a random sample from $U[0, \theta]$ distribution. Compare the mean squared errors of four estimators of θ : $T_1 = 5X_{1:4}$, $T_2 = (5/2)X_{2:4}$, $T_3 = (5/3)X_{3:4}$, and $T_4 = (5/4)X_{4:4}$.
- 11.3.9** Let $X_1, \dots, X_n$ be a random sample from the distribution with density $f(x, \theta) = 2x/\theta^2$ for $0 \leq x \leq \theta$ and 0 otherwise. Determine the MSE of $\hat{\theta} = X_{n:n}$.

11.4 EFFICIENCY

We assume that the observations $X_1, X_2, \dots$ are iid random variables, with $E(X_i) = \theta$ and $\text{Var}(X_i) = \sigma^2 < \infty$, the latter assumed known.

The obvious choice of estimator of the mean θ is $T = (X_1 + \dots + X_n)/n$, traditionally denoted by $\bar{X}$ or $\bar{X}_n$, if we need to stress the dependence on the sample size. Since

$$E(\bar{X}_n) = \frac{1}{n} E(X_1 + \dots + X_n) = \theta, \quad (11.26)$$

$\bar{X}_n$ is always an unbiased estimator of the mean θ . Furthermore, by the Chebyshev inequality and the fact that $\text{Var}(\bar{X}_n) = \sigma^2/n$ for every n ,

$$P\{|\bar{X}_n - \theta| > \epsilon\} \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \rightarrow 0, \quad (11.27)$$

and therefore $\bar{X}_n$ is a consistent estimator of θ if only $\text{Var}(X_i) < \infty$. In view of unbiasedness, the risk of $\bar{X}_n$ is equal to its variance, that is,

$$\text{MSE}_\theta(\bar{X}_n) = \frac{\sigma^2}{n}. \quad (11.28)$$

Let us now compare the situation with that of estimating the range θ of uniform distribution, by using estimators such as T_1 or T_2 given by (11.2) and (11.3). There are two main points of difference here. The first is that in the case of estimating the mean by $\bar{X}_n$, the risk for a fixed n is constant—it does not depend on the estimated parameter θ . In the case of estimators T_1 and T_2 , the risk given by (11.22) and (11.23) is a quadratic function of θ : the larger the value estimated is, the larger is the variance of the results.

The second difference—and much more important—is the rate at which the risk changes with the sample size n . In the case of estimating the mean by the sample average $\bar{X}_n$, the risk changes inversely with n , whereas in estimating the range of uniform distribution by T_1 , or T_2 , it changes (approximately) inversely with n^2 . Naturally, the latter case is more desirable practically: increasing the sample size by the factor of 10 reduces the MSE by factor of 100—that is, to about 1% of the MSE for the original sample size. As opposed to that, in estimating the mean by $\bar{X}_n$, an increase of sample size by the factor of 10 decreases the MSE by the factor of 10 only—that is, to about 10% of the MSE for the original sample size.

So why do statisticians use the sample average as an estimator, if there are other estimators that are so much better?

The answer is that a situation such as the one with the sample mean—when the MSE decreases inversely with n —is much more common than the situation where the MSE decreases with n^2 . In most cases, the best estimators that one can obtain have a MSE that decreases in proportion to $1/n$.

The concept that we will attempt to formalize is that of the average “amount of information about θ contained in a single observation of X .” Let $X_1, X_2, \dots$ be a random sample, selected from the distribution $f(x, \theta)$, where θ takes values in some open domain, and the set of points x at which $f(x, \theta) > 0$ does not depend on θ . (The last assumption rules out the case of observations from the $U[0, \theta]$ distribution, where the set of points at which $f(x, \theta)$ is positive depends on θ .) The reason for distinguishing such “regular” case is that if this assumption were violated, a single observation may eliminate some values of θ with certainty (e.g., in the case of uniform distribution on $(0, \theta]$, observation $X = 3$ rules out all $\theta < 3$). Such elimination is typically unattainable in practical situations; hence we concentrate the theory on “regular” cases.

Let us start from some intuitions. Suppose one gets a message that some event A occurred. The amount of information in this message depends, in an obvious way, on what event A is and who receives the message (think here of a message that Delta flight 503 will arrive 2 hours late). Apart from such semantic and personal information, there is also some information contained in the message (that A occurred) depending only on how likely is the event A . If $P(A)$ is close to 1, the amount of information is close to 0, while for $P(A)$ close to 0, the amount of information is high.

This interpretation agrees with common intuition, as well as with practice (e.g., in newspaper publishing). The fact that Mr. Smith found a dead roach in the can of beer he drank is of interest and worth reporting precisely because such an event is rare; hence, its occurrence carries lots of information. If, on the other hand, dead roaches were commonly found in cans of beer, no one would care to report it.

An interesting and coherent theory is obtained if the amount of information in the occurrence of an event with probability p is $\log(1/p) = -\log p$.

The situation in the case of estimation is somewhat more complicated, since we want to define the amount of information *about* θ in the event $X = x$, where X is a random variable with distribution (density or pdf) $f(x, \theta)$. Here $f(x, \theta)$ plays the role of probability p of the

event $X = x$. Since we are interested in information about θ , it appears natural to consider the rate of change of $\log(1/[f(x, \theta)]) = -\log f(x, \theta)$ under varying θ at the point x , that is, the derivative

$$\frac{\partial}{\partial \theta} [-\log f(x, \theta)] = -\frac{\frac{\partial}{\partial \theta} f(x, \theta)}{f(x, \theta)}. \quad (11.29)$$

We eliminate the effect of the sign by considering the square of quantity (11.29) and take the expectation to avoid the restriction to specific value x .

These considerations lead to the following definition:

Definition 11.4.1 Let X be a random variable with distribution $f(x, \theta)$, such that the set of points x at which $f(x, \theta) > 0$ is the same for all θ . We assume that the function $f(x, \theta)$ is twice differentiable with respect to θ for every x . Then, the *Fisher information* about θ in a single observation X is defined as

$$I(\theta) = E\{J(X, \theta)\}^2, \quad (11.30)$$

where

$$J(X, \theta) = \frac{\partial}{\partial \theta} [\log f(X, \theta)], \quad (11.31)$$

provided that the expectation (11.30) exists. Function $J(X, \theta)$ in (11.31) is called the *score function*. $\square$

Thus, in the case of a continuous random variable X , the quantity (11.30) is

$$I(\theta) = \int_{-\infty}^{+\infty} [J(x, \theta)]^2 f(x, \theta) dx = \int_{-\infty}^{+\infty} \left[\frac{\partial f(x, \theta)}{\partial \theta} \right]^2 \frac{1}{f(x, \theta)} dx,$$

while in the discrete case integration is replaced by summation.

■ EXAMPLE 11.15

Let X have normal distribution $N(\theta, \sigma^2)$. Then,

$$\log f(x, \theta) = -\log(\sigma\sqrt{2\pi}) - \frac{(x - \theta)^2}{2\sigma^2}$$

and

$$J(X, \theta) = \frac{\partial}{\partial \theta} [\log f(X, \theta)] = \frac{X - \theta}{\sigma^2}. \quad (11.32)$$

Thus,

$$I(\theta) = E\left(\frac{X - \theta}{\sigma^2}\right)^2 = \frac{1}{\sigma^4} E(X - \theta)^2 = \frac{\sigma^2}{\sigma^4} = \frac{1}{\sigma^2}.$$

■ EXAMPLE 11.16

Consider now a single Bernoulli trial with probability θ . We have $P\{X = 1|\theta\} = \theta$ and $P\{X = 0|\theta\} = 1 - \theta$ so that

$$f(x, \theta) = \theta^x (1 - \theta)^{1-x}, \quad x = 0, 1. \quad (11.33)$$

Thus, $\log f(x, \theta) = x \log \theta + (1 - x) \log(1 - \theta)$ and

$$\left[\frac{\partial}{\partial \theta} \log f(x, \theta) \right]^2 = \left(\frac{x}{\theta} - \frac{1-x}{1-\theta} \right)^2 = \begin{cases} \frac{1}{(1-\theta)^2} & \text{if } x = 0 \\ \frac{1}{\theta^2} & \text{if } x = 1. \end{cases} \quad (11.34)$$

Taking the expectation, we obtain

$$I(\theta) = \frac{1}{(1-\theta)^2} \times (1-\theta) + \frac{1}{\theta^2} \times \theta = \frac{1}{\theta(1-\theta)}. \quad (11.35)$$

■ EXAMPLE 11.17

Let us slightly modify Example 11.16: Suppose now that $P\{X = 1|\theta\} = \theta^2$ and $P\{X = 0|\theta\} = 1 - \theta^2$; thus, we have a single Bernoulli trial, but now the probability of success is θ^2 . However, we are still interested in the amount of information about θ (not θ^2). A practical example here may be found in genetics, where θ is a frequency of a recessive gene (e.g., a gene causing a person to have blue eyes) so that $X = 1$ corresponds to finding a person with some features caused by a recessive gene (requiring both parents to transmit this gene). We now have

$$f(x, \theta) = (\theta^2)^x (1 - \theta^2)^{1-x};$$

hence

$$\begin{aligned} \left[\frac{\partial}{\partial \theta} \log f(x, \theta) \right]^2 &= \left\{ \frac{\partial}{\partial \theta} [2x \log \theta + (1-x) \log(1 - \theta^2)] \right\}^2 \\ &= \left\{ \frac{2x}{\theta} - 2\theta \frac{(1-x)}{1-\theta^2} \right\}^2 \\ &= \begin{cases} \frac{4\theta^2}{(1-\theta^2)^2} & \text{if } x = 0 \\ \frac{4}{\theta^2} & \text{if } x = 1. \end{cases} \end{aligned}$$

Consequently,

$$I(\theta) = \frac{4\theta^2}{(1-\theta^2)^2} (1-\theta^2) + \frac{4}{\theta^2} \theta^2 = \frac{4\theta^2}{1-\theta^2} + 4 = \frac{4}{1-\theta^2}. \quad (11.36)$$

Comparison of this result with the result in Example 11.16 is quite instructive. In a single Bernoulli trial with probability of success θ , the average amount of information in a trial is a function that assumes its minimal value 4 at $\theta = 1/2$. As θ moves away from $1/2$ toward either $\theta \approx 0$ (success very rare) or $\theta = 1$ (failure very rare), the average amount of information about θ increases to infinity.

On the other hand, if we can only observe whether or not an event with probability θ^2 has occurred, then the average amount of information about θ is close to 4 if $\theta \approx 0$, and increases to infinity when $\theta \rightarrow 1$. The symmetry of the preceding example is lost. Indeed, only when θ is close to 1 is θ^2 also close to 1, so successes that are sufficiently frequent provide a good estimator of θ^2 , and hence also of θ . For small θ , we have a poor estimate of θ^2 , and hence also a poor estimate of θ .

Alternative formulas for $I(\theta)$ can also be derived. Observe first that

$$\begin{aligned} E_\theta[J(X, \theta)] &= \int_{-\infty}^{\infty} \left[\frac{\partial}{\partial \theta} \log f(x, \theta) \right] f(x, \theta) dx \\ &= \int_{-\infty}^{\infty} \left[\frac{\frac{\partial}{\partial \theta} f(x, \theta)}{f(x, \theta)} \right] f(x, \theta) dx = \int_{-\infty}^{\infty} \frac{\partial}{\partial \theta} f(x, \theta) dx \\ &= \frac{\partial}{\partial \theta} \int_{-\infty}^{\infty} f(x, \theta) dx = \frac{\partial}{\partial \theta} (1) = 0. \end{aligned} \quad (11.37)$$

This is valid under the conditions that allow to interchange differentiation with respect to θ and integration with respect to x . Thus, in view of (11.37),

$$\text{Var}_\theta[J(X, \theta)] = E_\theta[J(X, \theta)]^2 - \{E[J(X, \theta)]\}^2 = I(\theta). \quad (11.38)$$

Still another formula for $I(\theta)$ can be obtained by noting that

$$\begin{aligned} \frac{\partial}{\partial \theta} J(x, \theta) &= \frac{\partial}{\partial \theta} \left[\frac{\frac{\partial}{\partial \theta} f(x, \theta)}{f(x, \theta)} \right] = \frac{\left(\frac{\partial^2}{\partial \theta^2} f(x, \theta) \right) - \left(\frac{\partial}{\partial \theta} f(x, \theta) \right)^2}{[f(x, \theta)]^2} \\ &= \frac{\frac{\partial^2}{\partial \theta^2} f(x, \theta)}{f(x, \theta)} - [J(x, \theta)]^2. \end{aligned}$$

Taking expectations of both sides, we get

$$E_\theta \left(\frac{\partial}{\partial \theta} J(x, \theta) \right) = \int_{-\infty}^{\infty} \frac{\partial^2}{\partial \theta^2} f(x, \theta) dx - E_\theta[J(X, \theta)]^2 = -I(\theta),$$

since, assuming one can interchange integration and differentiation,

$$\int_{-\infty}^{\infty} \frac{\partial^2}{\partial \theta^2} f(x, \theta) dx = \frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{\infty} f(x, \theta) dx = \frac{\partial^2}{\partial \theta^2} (1) = 0.$$

We have therefore the following theorem:

Theorem 11.4.1 *If the density $f(x, \theta)$ is twice differentiable in θ and the equality $\int f(x, \theta) dx = 1$ can be differentiated twice under the integral sign, we have*

$$I(\theta) = E_\theta[J(X, \theta)]^2 = \text{Var}_\theta[J(X, \theta)] = -E_\theta \left(\frac{\partial}{\partial \theta} J(x, \theta) \right). \quad (11.39)$$

□

The following theorem determines the average information contained in a random sample of size n :

Theorem 11.4.2 *The information $I_n(\theta)$, in a random sample of n observations, is n times the information of a single observation:*

$$I_n(\theta) = nI(\theta). \quad (11.40)$$

Proof: The density of $\mathbf{X}$ is $f(\mathbf{X}, \theta) = f(X_1, \theta) \cdots f(X_n, \theta)$ so that

$$J(\mathbf{X}, \theta) = \frac{\partial}{\partial \theta} \log f(\mathbf{X}, \theta) = \frac{\partial}{\partial \theta} \sum_{i=1}^n \log f(X_i, \theta) = \sum_{i=1}^n J(X_i, \theta).$$

Differentiating both sides, we have

$$\frac{\partial}{\partial \theta} J(\mathbf{X}, \theta) = \sum_{i=1}^n \frac{\partial}{\partial \theta} J(X_i, \theta). \quad (11.41)$$

□

The next theorem connects the amount of information about θ contained in a single observation, with the variance of an estimator of θ .

Theorem 11.4.3 Cramér–Rao Inequality *For any estimator T_n such that $E_\theta(T_n) = m(\theta)$, we have*

$$\text{Var}_\theta(T_n) \geq \frac{[m'(\theta)]^2}{nI(\theta)}. \quad (11.42)$$

In particular, if T is an unbiased estimator of θ , then $m(\theta) = \theta$, $m'(\theta) = 1$, and

$$\text{Var}_\theta(T_n) \geq \frac{1}{nI(\theta)}. \quad (11.43)$$

Proof: Let $T(\mathbf{X}) = T(X_1, \dots, X_n)$ be an arbitrary estimator of θ , assumed to have a finite variance. Letting $\mathbf{x} = (x_1, \dots, x_n)$, we have

$$m(\theta) = E_\theta(T) = \int \cdots \int T(\mathbf{x}) f(\mathbf{x}, \theta) dx_1 \cdots dx_n.$$

Consequently, assuming again that we can differentiate under the integral sign, we obtain

$$\begin{aligned} m'(\theta) &= \int \cdots \int T(\mathbf{x}) \left[\frac{\partial}{\partial \theta} f(\mathbf{x}, \theta) \right] dx_1 \cdots dx_n \\ &= \int \cdots \int T(\mathbf{x}) \left[\frac{\frac{\partial}{\partial \theta} f(\mathbf{x}, \theta)}{f(\mathbf{x}, \theta)} \right] f(\mathbf{x}, \theta) dx_1 \cdots dx_n \\ &= \int \cdots \int T(\mathbf{x}) J(\mathbf{x}, \theta) f(\mathbf{x}, \theta) dx_1 \cdots dx_n \\ &= E_\theta\{T(\mathbf{X})J(\mathbf{X}, \theta)\} = \text{Cov}_\theta\{T(\mathbf{X}), J(\mathbf{X}, \theta)\}. \end{aligned} \quad (11.44)$$

The last equality follows from the fact that $E[J(\mathbf{X}, \theta)] = 0$, which can be established in the same way as for a single random variable X , in view of the relation

$$J(\mathbf{X}, \theta) = \sum_{i=1}^n J(X_i, \theta).$$

Now, by the Schwarz inequality (7.6.7) and (11.44), we may write

$$[m'(\theta)]^2 = [\text{Cov}_\theta(T(\mathbf{X}), J(\mathbf{X}, \theta))]^2 \leq \text{Var}_\theta(T) \times \text{Var}_\theta[J(\mathbf{X}, \theta)]. \quad (11.45)$$

□

Observe that we have equality in (11.42) if and only if we have equality in (11.45), which means that the correlation coefficient between $T(X)$ and $J(X, \theta)$ satisfies the condition $|\rho_{T(X), J(X, \theta)}| = 1$. By Theorem 7.5.6, this is equivalent to linear relationship between $T(X)$ and $J(X, \theta)$, that is,

$$T(X) = \gamma_1(\theta)J(X, \theta) + \gamma_2(\theta) \quad (11.46)$$

for some functions $\gamma_1(\theta)$ and $\gamma_2(\theta)$.

An important point here is that T is an estimator of θ , so the left-hand side does not depend on θ . Consequently, the right-hand side does not depend on θ either; that is, functions

γ_1 and γ_2 cancel the dependence of $J(X, \theta)$ on the parameter θ . The set of conditions that allow all differentiations under the integral sign, combined with the condition of independence of θ of the set of x for which $f(x, \theta) > 0$, will be called the regularity conditions.

We have the following definition:

Definition 11.4.2 Any unbiased estimator T that satisfies the regularity conditions and whose variance attains the minimum equal to the right-hand side of (11.43) is called *efficient*. The ratio $nI(\theta)/\text{Var}_\theta(T)$ is called the *efficiency* of T .

More generally, given two unbiased estimators T_1 and T_2 of θ , the ratio of their variances $\text{Var}_\theta(T_1)/\text{Var}_\theta(T_2)$ is called the *relative efficiency* of T_2 with respect to T_1 . $\square$

■ EXAMPLE 11.18

Consider the problem of estimating the mean θ of normal distribution $N(\theta, \sigma^2)$. From Example 11.15, we have $I(\theta) = 1/\sigma^2$. The most obvious estimator for θ , $\bar{X}_n$, is unbiased, and $\text{Var}(\bar{X}_n) = \sigma^2/n$. The lower bound of variances of all unbiased estimators of θ , based on samples of size n , is $1/[nI(\theta)] = \sigma^2/n$, which shows that $\bar{X}_n$ is efficient.

■ EXAMPLE 11.19

Continuing Example 11.16, suppose that we want to estimate probability of success θ on the basis of n Bernoulli trials. Since $I(\theta) = 1/[\theta(1-\theta)]$, the information in the sample of size n is $n/[\theta(1-\theta)]$. For $S_n = X_1 + \cdots + X_n$ being the total number of successes, $T = S_n/n$ will be an estimator of θ . S has binomial distribution; $E(T) = (1/n)E(S_n) = \theta$, and $\text{Var}(T) = (1/n^2)\text{Var}(S_n) = \theta(1-\theta)/n$, which shows that T is efficient.

■ EXAMPLE 11.20

In Section 11.2, we studied five estimators of θ in $U[0, \theta]$ distribution. We found that variances of estimators T_1, T_2 , and T_3 decrease to zero like $1/n^2$ (and not at the rate $1/n$, as for efficient estimators). Thus, those estimators are superefficient. This is possible since $f(x, \theta)$ does not satisfy the regularity conditions. The set of points x at which the density $f(x, \theta)$ is positive is $[0, \theta]$; hence it depends on θ .

Under the regularity conditions, the right-hand side of the Cramér–Rao inequality (11.43) gives the lower bound for variances of unbiased estimators, and therefore, if an efficient estimator of θ is found, it is known that a better unbiased estimator does not exist. It is necessary to point out, however, that the Cramér–Rao bound is not always attainable. It may happen that the best possible unbiased estimator has a variance larger than the Cramér–Rao bound. For a thorough discussion of these topics see, for example, Mood et al. (1974).

PROBLEMS

11.4.1 Let $\bar{X}_{2k}$ be the sample mean of $2k$ independent observations from a normal distribution with mean θ and known variance σ^2 . Find the efficiency of $\bar{X}_k$ (i.e., of estimator that uses only half of the sample).

11.4.2 Let X have $\text{EXP}(\lambda)$ distribution. Find the Fisher information $I(\lambda)$.

11.4.3 Let $X_1, \dots, X_n$ be a random sample from $\text{EXP}(\lambda)$ distribution. Propose an efficient estimator of $1/\lambda$ and determine its variance.

- 11.4.4** Find, if it exists, the Fisher information $I(\theta)$ in a random sample of size n from:
(i) The Cauchy distribution with density $f(x, \theta) = \{\pi[1 + (x - \theta)^2]\}^{-1}$, $-\infty < x < \infty$. **(ii)** The Rayleigh distribution with density $f(x, \theta) = (x/\theta) \exp\{-x^2/(2\theta)\}$, $x \geq 0, \theta > 0$. **(iii)** The Gumbel generalized extreme value distribution with density $f(x, \theta) = \exp\{-(x - \theta) + \exp\{-(x - \theta)\}\}$, $-\infty < x < \infty, -\infty < \theta < \infty$.
- 11.4.5** Let X_1, X_2 be a random sample of size 2 from $N(\mu, \sigma^2)$ distribution. Determine the amount of information about μ and about σ^2 contained in: **(i)** $X_1 + X_2$. **(ii)** $X_1 - X_2$.
- 11.4.6** Let $X_1, \dots, X_n$ be a random sample from a Bernoulli distribution with an unknown p . Show that the variance of any unbiased estimator of $(1 - p)^2$ must be at least $4p(1 - p)^3/n$.
- 11.4.7** Show that the estimator $T = \bar{X}$ satisfies relation (11.46) and determine functions γ_1 and γ_2 if a random sample $X_1, X_2, \dots, X_n$ is selected from: **(i)** $N(\theta, \sigma^2)$ distribution with σ known. **(ii)** $\text{BIN}(1, \theta)$ distribution.
- 11.4.8** Let $X_1, \dots, X_n$ be a random sample from the $\text{POI}(\lambda)$ distribution. Find the Cramér–Rao lower bound for variances of unbiased estimators of $P(X \leq 1) = (1 + \lambda)e^{-\lambda}$.

11.5 METHODS OF OBTAINING ESTIMATORS

Before introducing further criteria for evaluating performance of estimators (besides consistency, unbiasedness, and efficiency), we shall now present methods of constructing estimators.

Method of Moments Estimators

In this approach, estimators are obtained by equating sample quantities to the corresponding population quantities as functions of the parameter to be estimated. The estimator is then found as a solution of the resulting equation with respect to the parameter, and the general approach is also known as a *plug-in principle*.

Since the most commonly taken quantity is a moment of the random variable, this method of constructing estimators is generally known as *method of moments*.

■ EXAMPLE 11.21

Let $X_1, \dots, X_n$ be a random sample from the $\text{EXP}(\theta)$ distribution. Then, $E(X_i) = 1/\theta$. The sample counterpart of the first moment is the sample mean $\bar{X}_n = (1/n)(X_1 + \dots + X_n)$. Equating the empirical and theoretical mean, we obtain the equation $\bar{X}_n = 1/\hat{\theta}$, which gives the estimator $T_1 = 1/\bar{X}_n$ of the parameter θ .

We could, however, use the second moment as well. Then we have $E(X^2) = \text{Var}(X) + [E(X)]^2 = 2/\theta^2$. The empirical counterpart of the second moment is $(1/n) \sum_{i=1}^n X_i^2$. We therefore obtain the equation

$$\frac{1}{n} \sum_{i=1}^n X_i^2 = \frac{2}{\hat{\theta}^2},$$

which, solved for $\hat{\theta}$, gives an estimator

$$T_2 = \sqrt{\frac{2n}{\sum_{i=1}^n X_i^2}}. \quad (11.47)$$

Still another possibility would be to use the median m , the point at which cdf equals $1/2$, so $F(m) = 1 - e^{-\theta m} = 1/2$, implying $m = (\log 2)/\theta$. The empirical counterpart of the median m is the sample median. We will assume for simplicity, that the sample size is odd ($n = 2k + 1$). Then the sample median is the $(k + 1)$ st order statistic $X_{k+1:n}$. We therefore have still another estimator of θ ,

$$T_3 = \frac{\log 2}{X_{k+1:n}}. \quad (11.48)$$

Finally, suppose that we want to estimate the probability

$$p = P\{X \geq 3\} = e^{-3\theta}.$$

The method of moments suggests using the estimator e^{-3T} , where T is any estimator of θ . Estimators T_1, T_2 , and T_3 now give three estimators of p , namely

$$\hat{p}_1 = \exp\left\{-\frac{3}{\bar{X}_n}\right\}, \quad \hat{p}_2 = \exp\left\{-3\sqrt{\frac{2n}{\sum X_i^2}}\right\}, \quad \hat{p}_3 = \exp\left\{\frac{-3 \log 2}{X_{k+1:n}}\right\}.$$

■ EXAMPLE 11.22

Let us observe that the estimator $T_5 = 2\bar{X}_n$ from Example 11.4 is also an example of estimator obtained by the method of moments. Since $E(X) = \theta/2$, doubling the sample mean gives a moment estimator of θ .

The method of moments can also be applied in the case of estimating several parameters at once, as illustrated by the following example:

■ EXAMPLE 11.23

Suppose that we want to estimate both μ and σ^2 based on a random sample $X_1, \dots, X_n$ from some distribution with mean μ and variance σ^2 .

SOLUTION. We could choose the following two expressions:

$$E(X) = \mu, \quad E(X^2) = \sigma^2 + \mu^2. \quad (11.49)$$

This suggests that we compare the first and second empirical moment with μ and $\sigma^2 + \mu^2$ by obtaining the equations

$$\frac{1}{n} \sum_{i=1}^n X_i = \hat{\mu}, \quad \frac{1}{n} \sum_{i=1}^n X_i^2 = \hat{\sigma}^2 + \hat{\mu}^2.$$

Solving for $\hat{\mu}$ and $\hat{\sigma}^2$, we obtain

$$\hat{\mu} = \bar{X} \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2. \quad (11.50)$$

Thus the sample mean and the sample variance (with divisor n) are method-of-moment estimators of the population mean and variance in the general case.

■ EXAMPLE 11.24

As the last example of the method-of-moment estimator, consider the problem of the randomized response, discussed in Section 1.2. Generally, the empirical problem concerns some attribute, call it Q , that a person might be reluctant to admit having. The objective is to estimate the frequency θ of persons in a given population whose true reply to a question “Are you a Q -person?” is “yes.” To collect the data on such questions one can use a questionnaire with a randomized response (see Example 1.9). The respondent activates two random mechanisms, one generating the event A or its complement A^c , the other generating the event B or its complement B^c . These two mechanisms operate independently. The probabilities $P(A) = \alpha$ and $P(B) = \beta$ are known; however, only the respondent knows which of the events A and B occurred in a given instance. He is instructed to respond to the question “Are you a Q -person?” if A occurs, and otherwise respond to the question “Did B occur?” The answer “yes” or “not” is recorded by the experimenter who does not know which question was actually answered.

Conditioning on the occurrence or nonoccurrence of event A , we have

$$\begin{aligned} P(\text{“yes”}) &= P(\text{“yes”}|A)P(A) + P(\text{“yes”}|A^c)P(A^c) \\ &= \theta\alpha + \beta(1 - \alpha). \end{aligned}$$

If now X respondents out of N tested replied “yes,” then X/N is an estimator of $P(\text{“yes”})$, and we have an approximate equality

$$\frac{X}{N} \approx \theta\alpha + \beta(1 - \alpha),$$

which suggests using as an estimator of θ the random variable

$$T = \frac{X/N - \beta(1 - \alpha)}{\alpha}.$$

A problem with the estimators obtained by method of moments is that they are not unique, since one can take moments of various orders, or even different quantities. However, as a rule, one should take the moments of lowest orders that depend on θ , as illustrated by the following example:

■ EXAMPLE 11.25

Suppose that the observations are known to have $U[-\theta, \theta]$ distribution. Then $E(X) = 0$, and the first moment contains no information about θ . One may use here the fact that

$$E(X^2) = \int_{-\theta}^{\theta} x^2 \times \frac{1}{2\theta} dx = \frac{\theta^2}{3},$$

which gives the estimator

$$\hat{\theta} = \sqrt{\frac{3}{n} \sum_{i=1}^n X_i^2}.$$

The moment estimators are consistent under some very mild conditions. Indeed, the strong law of large numbers for the iid case asserts that $(1/n)(X_1^k + X_2^k + \cdots + X_n^k)$

converges with probability 1 to $E_\theta(X^k)$ if only $E_\theta(|X|^k) < \infty$. Consequently, the empirical moments converge also in probability to the corresponding theoretical moments, and if only the parameter θ is a continuous function of moments (as is usually the case), the consistency of method-of-moment estimators follows.

The situation is not so straightforward for estimators built on quantities other than the moments. Consistency has to be studied separately in each such case, by establishing whether or not the sample analogue of a given quantity converges in probability (or almost surely) to the corresponding theoretical quantity. However, the main issue with method-of-moment estimators is that they either coincide with estimators obtained by the maximum likelihood method (discussed below), or they are inferior to them.

Maximum Likelihood Estimators

Let $X_1, \dots, X_n$ be a random sample from distribution $f(x, \theta)$, where $f(x, \theta)$ may stand for the density or for the probability function. If the observed values are $X_1 = x_1, \dots, X_n = x_n$, then the probability of this sample (or the joint density) is

$$f(x_1, \theta) \times \cdots \times f(x_n, \theta). \quad (11.51)$$

The product (11.51), regarded as a function of parameter θ , is called the *likelihood function* of the sample, or simply the likelihood function. We will use the symbol

$$L(\theta) = L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n f(x_i, \theta) = f_n(\mathbf{x}, \theta), \quad (11.52)$$

where $\mathbf{x} = (x_1, \dots, x_n)$.

Definition 11.5.1 Given the sample $\mathbf{x} = (x_1, \dots, x_n)$, the value $\hat{\theta} = \hat{\theta}(\mathbf{x})$ that maximizes the likelihood function (11.52), is called the *maximum likelihood estimate* (MLE) of θ . $\square$

Let us begin with some examples:

■ EXAMPLE 11.26

We will find MLE of the probability of success θ , if in five independent Bernoulli trials with probability of success θ , three successes and two failures were observed.

We have here $f(x, \theta) = \theta^x(1 - \theta)^{1-x}$ where $x = 0$ or 1 represent failure and success, respectively. Thus the likelihood is

$$L(\theta) = \prod_{i=1}^5 \theta^{x_i}(1 - \theta)^{1-x_i} = \theta^3(1 - \theta)^2, \quad (11.53)$$

where $0 \leq \theta \leq 1$. The information as to which trials led to successes and which to failures does not affect the likelihood and is irrelevant for the estimation of θ .

To find the maximum of function (11.53) we differentiate, obtaining the equation

$$\begin{aligned} L'(\theta) &= 3\theta^2(1 - \theta)^2 - 2\theta^3(1 - \theta) = \theta^2(1 - \theta)[3(1 - \theta) - 2\theta] \\ &= \theta^2(1 - \theta)(3 - 5\theta). \end{aligned}$$

The solutions of equation $\theta^2(1 - \theta)(3 - 5\theta) = 0$ are: $\theta = 0$, $\theta = 1$, and $\theta = 3/5$. An inspection of $L(\theta)$ shows that this function attains its maximum at the last solution, while $\theta = 0$ and $\theta = 1$ give the minima. Thus the MLE of θ is $\hat{\theta} = 3/5$.

■ **EXAMPLE 11.27**

Suppose that we take $n = 3$ observations from $\text{POI}(\theta)$ distribution, obtaining values $x_1 = 2, x_2 = 0, x_3 = 5$. The likelihood is

$$L(\theta) = \left(\frac{\theta^2}{2!}e^{-\theta}\right) \left(\frac{\theta^0}{0!}e^{-\theta}\right) \left(\frac{\theta^5}{5!}e^{-\theta}\right) = \frac{\theta^7}{2!5!}e^{-3\theta}.$$

The derivative now is

$$L'(\theta) = \frac{1}{2!5!}(7\theta^6e^{-3\theta} - 3\theta^7e^{-3\theta}),$$

and $L'(\theta) = 0$ for $\theta = 0$ and for $\theta = 7/3$. An inspection of L shows that the maximum occurs at the second solution, so the MLE of θ is now $7/3$.

■ **EXAMPLE 11.28**

Suppose that we observe values $x_1 = 3$ and $x_2 = -2$ from a $N(0, \theta^2)$ distribution. The likelihood now is

$$L(\theta) = \frac{1}{\theta\sqrt{2\pi}}e^{-9/2\theta^2} \times \frac{1}{\theta\sqrt{2\pi}}e^{-4/2\theta^2} = \frac{1}{2\pi\theta^2}e^{-13/2\theta^2}.$$

Since $L(\theta)$ is maximized at the same point at which its logarithm is maximized, we will first take the logarithm and then differentiate. Taking natural logarithms, we have

$$\log L(\theta) = -2\log \theta - \log(2\pi) - \frac{13}{2\theta^2}.$$

Hence

$$\frac{d}{d\theta} \log L(\theta) = -\frac{2}{\theta} + \frac{13}{\theta^3},$$

and we obtain the equation

$$-\frac{1}{\theta} \left(2 - \frac{13}{\theta^2}\right) = 0$$

with solution $\theta = \sqrt{13/2}$. After checking that the likelihood is maximized at $\sqrt{13/2}$, we obtain $\hat{\theta} = \sqrt{13/2}$ as the MLE of θ .

Some obvious questions arise here. First, the *statistical* questions concerning the properties of the suggested procedure. The point is that the MLE depends on the sample, so it is a random quantity. Our procedure therefore defines an estimator. What are its properties, such as consistency, bias and efficiency?

Second, the *mathematical* questions: Does MLE always exist? If so, is the maximum of the likelihood function unique? Can it always be obtained by differentiation of likelihood or of its logarithm and solving the resulting equation?

There are cases when, formally speaking, MLE does not exist, but these cases can often be modified in a natural way so as to remedy the situation.

■ **EXAMPLE 11.29**

Consider the problem of estimating θ based on observations from the $U(0, \theta)$ distribution, with density

$$f(x, \theta) = \begin{cases} 1/\theta & \text{for } 0 < x < \theta \\ 0 & \text{otherwise.} \end{cases}$$

Given the sample $x_1, \dots, x_n$, the likelihood is

$$L(\theta) = f(x_1, \theta) \cdots f(x_n, \theta) = \begin{cases} 1/\theta^n & \text{if } 0 < x_i < \theta \text{ for all } i \\ 0 & \text{otherwise.} \end{cases}$$

Thus the likelihood function is discontinuous: it is the function depicted in Figure 11.1, with discontinuity at the point $t = \max(x_1, \dots, x_n)$, and decreasing for $\theta > t$. However, for $\theta = t$ we have $L(t) = 0$, so there is no point at which this function attains its maximum, and MLE does not exist.

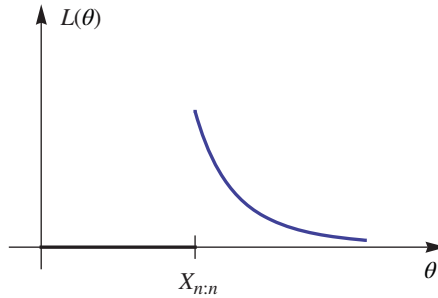


Figure 11.1 Likelihood function for the range R in uniform distribution.

The cause of the trouble here is the choice of the definition of the density. If we define $f(x, \theta)$ to be $1/\theta$ in the *closed* interval $0 \leq x \leq \theta$, the likelihood would actually reach its maximum at $\hat{\theta} = \max(x_1, \dots, x_n)$.

The example above shows a type of situation where MLE does not exist because of the choice of the density function. Since the density function can be modified at a single point (or at any finite or countable set of points), such a modification can affect the likelihood function. Consequently, we may (and will) always assume that the densities are defined in such a way that the maximum of the likelihood exists.

■ EXAMPLE 11.30

Assume that we are estimating θ based on a random sample $X_1, \dots, X_n$, with X_i 's being distributed uniformly in the interval $[\theta - 1/2, \theta + 1/2]$. We have

$$f(x, \theta) = \begin{cases} 1 & \text{for } \theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2} \\ 0 & \text{otherwise.} \end{cases}$$

Consequently, $L(\theta) = 1$ if $X_{n:n} \leq \theta + 1/2$ and $X_{1:n} \geq \theta - 1/2$, and is equal 0 otherwise. All values of $L(\theta)$ in a certain interval are equal 1, and the maximum is not unique. All values between $X_{n:n} - 1/2$ and $X_{1:n} + 1/2$ are MLE's of θ .

Despite the above-mentioned shortcomings, MLEs are reasonable in most cases appearing in practice. We will continue with the example of the MLE in the case of a fixed number of Bernoulli trials.

EXAMPLE 11.31

Let us consider, as in Example 11.26, five Bernoulli trials with probability of success θ . The information on which trials led to success and which led to failures is not essential: what matters is the total number S of successes. The likelihood function for $S = s$ is

$$L(\theta, s) = \theta^s (1 - \theta)^{5-s}.$$

For $0 \leq \theta \leq 1$, the function $L(\theta, s)$ is maximized at $s/5$ if $s = 1, 2, 3$, or 4 , as can be seen by taking derivatives and solving the equation $L'(\theta) = 0$. For $s = 0$, we have $L(\theta, 0) = (1 - \theta)^5$, and the maximum occurs at $\theta = 0$. Similarly, for $s = 5$, we have $L(\theta, 5) = \theta^5$, and the maximum occurs at $\theta = 1$. Thus the MLE of θ , for given s , is $s/5$ (although for $s = 0$ and $s = 5$ the maximum occurs at the boundary and therefore cannot be established by taking derivatives).

Graphically the six likelihood functions corresponding to various numbers s of successes are presented in Figure 11.2. If we now regard s as a value of random variable S , then the likelihood function $L(\theta, S)$ becomes random. The corresponding point at which $L(\theta, 5)$ attains its maximum equal to $S/5$, is also random. Thus the MLE becomes a random variable, dependent on the sample $(X_1, \dots, X_n)$ through the statistic $S = X_1 + \dots + X_5$.

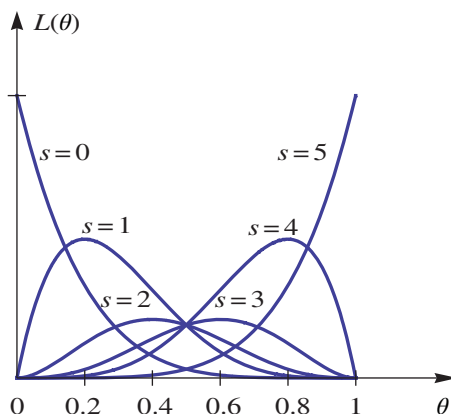


Figure 11.2 Likelihood function for five Bernoulli trials.

We will regard the likelihood function as a random function of θ , the randomness being induced by the sample $X_1, \dots, X_n$. The value $\hat{\theta}$ that maximizes the likelihood will be called a *maximum likelihood estimator* of θ . Following tradition, we will use the same MLE symbol for both the maximum likelihood estimator (random variable) and for the maximum likelihood estimate (its value for a particular sample).

Let us begin by stating some properties of maximum likelihood estimators. The property that makes these estimators quite convenient is known as *invariance*.

Invariance Principle *If $\hat{\theta}$ is the MLE of parameter θ , then $h(\hat{\theta})$ is the MLE of parameter $h(\theta)$.*

Let us begin with the case where the mapping h is one-to-one. Then there exists the inverse mapping g of Θ' onto Θ such that $\theta' = h(\theta)$ whenever $\theta = g(\theta')$. If the likelihood $L(\theta, \mathbf{x})$ is maximized at the point $\hat{\theta} = \hat{\theta}(\mathbf{x})$, then the function $L(g(\theta'), \mathbf{x})$ is maximized when $g(\theta') = \hat{\theta}(\mathbf{x})$, and hence when $\theta' = h[\hat{\theta}(\mathbf{x})]$. Perhaps the most common application of the invariance

principle is the fact that if $\hat{\theta}$ is the MLE of the variance σ^2 , then $\sqrt{\hat{\theta}}$ is the MLE of the standard deviation σ .

The invariance principle is valid in the case of multidimensional parameters, and one-dimensional functions of such parameters. Consider the following example:

■ EXAMPLE 11.32

Suppose that the sample is taken from a distribution $f(x; \mu, \sigma)$, where μ and σ are the mean and standard deviation (we may consider f as normal, but it is not necessary). Thus $\theta = (\mu, \sigma)$ is a two-dimensional parameter. Assume that the parameter space is $\Theta = \{\mu > 0, \sigma > 0\}$. Suppose that we want to estimate the coefficient of variation $\nu = \sigma/\mu$.

The invariance principle asserts that if $\hat{\mu}$ and $\hat{\sigma}$ are MLE's of μ and σ , then $\hat{\nu} = \hat{\sigma}/\hat{\mu}$ is the MLE of the coefficient of variation ν . It should be recognized, however, that this conclusion does not follow from previous reasoning, since the function that maps $\theta = (\mu, \sigma)$ into $h(\theta) = \sigma/\mu$ is not one-to-one.

The argument showing that the invariance principle for the MLE's is also valid in the multidimensional case is as follows: Assume that $\theta = (\eta_1, \eta_2, \dots, \eta_m)$ is an m -dimensional parameter, and let $h(\theta) = h(\eta_1, \eta_2, \dots, \eta_m)$ be a function of θ to be estimated. Then find $m-1$ functions $h_2(\theta) = h_2(\eta_1, \eta_2, \dots, \eta_m), \dots, h_m(\theta) = h_m(\eta_1, \eta_2, \dots, \eta_m)$ such that the vector

$$H(\theta) = (h(\theta), h_2(\theta), \dots, h_m(\theta))$$

is a one-to-one mapping of m -dimensional parameter space Θ into a subset Θ' of the m -dimensional space. By the preceding argument, if $\hat{\theta} = (\hat{\eta}_1, \dots, \hat{\eta}_m)$ is the MLE of θ , then

$$H(\hat{\theta}) = (h(\hat{\eta}_1, \dots, \hat{\eta}_m), h_2(\hat{\eta}_1, \dots, \hat{\eta}_m), \dots, h_m(\hat{\eta}_1, \dots, \hat{\eta}_m))$$

is the MLE of $H(\theta)$. It follows therefore that $h(\hat{\eta}_1, \dots, \hat{\eta}_m) = h(\hat{\theta})$ is the MLE of $h(\theta)$.

The second important property of maximum likelihood estimators is that they do not depend on the design of the experiment. To explain the issues involved here, we start by formulating the likelihood principle.

Likelihood Principle Consider two sets of data, $\mathbf{x}$ and $\mathbf{y}$, obtained from the same population, although possibly according to different sampling plans. If the ratio of their likelihoods, $L_1(\theta, \mathbf{x})/L_2(\theta, \mathbf{y})$, does not depend on θ , then both data sets provide the same information about the parameter θ and consequently should lead to the same conclusion about θ .

Consequently, what matters are the ratios of likelihoods rather than the likelihoods themselves. We will explain by some examples how this principle makes the MLE's independent on the design of experiment.

■ EXAMPLE 11.33

Assume that we want to estimate θ , the probability of success in Bernoulli trials. One experimental design consists of fixing the number n of observations and recording the number of successes. If we observe x successes, the likelihood is $L_1(\theta, x) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$.

Suppose now that one decides to fix x and take observations until x successes are recorded. Now the probability that the observations will end on the n th trial is given by negative binomial distribution, so the likelihood is

$$L_2(\theta, n) = \binom{n-1}{x-1} \theta^x (1-\theta)^{n-x}.$$

Observe that in the first case x was random and n fixed; in the second it is the other way around. Nevertheless, the ratio of these two likelihoods is $\binom{n}{x} / \binom{n-1}{x-1}$, which does not depend on θ . The MLE of θ is $\hat{\theta} = x/n$ in either case. Therefore the additional information that in the second case the last experiment led to success does not affect our estimate of θ .

■ EXAMPLE 11.34

Consider tennis players A and B who from time to time play a match against each other. Let us assume that the probability of A winning a set against B is p , and the results of sets are independent (This is a somewhat oversimplified assumption; probability p may change over time, and results of sets within a match may be dependent. We will, however, take this assumption as a starting point for analysis.) We want to estimate the probability p , which reflects the relative strengths of players A and B .

SOLUTION. Assume that A and B are men.²³ Therefore the probability that A will win in 3 sets is $\alpha_3 = p^3$. The probability that he will win in 4 sets is $\alpha_4 = 3p^3(1-p)$, since he must win the last set (probability p) and two of the first three sets [probability $\binom{3}{2}p^2(1-p)$]. Similarly the probability of winning in 5 sets is $\alpha_5 = 6p^3(1-p)^2$. The analogous probabilities of winning by player B in 3, 4, and 5 sets are β_3, β_4 , and β_5 , obtained by interchanging the roles of p and $1-p$.

Now assume that the sport section of a newspaper provides data for the last year, in the form of six numbers $(a_3, a_4, a_5, b_3, b_4, b_5)$, where a_i is the number of matches won by A (against B) in i sets, and similarly for b_i . Letting $q = 1-p$, the likelihood of the data is

$$\begin{aligned} L(p; a_3, \dots, b_5) &= \alpha_3^{a_3} \alpha_4^{a_4} \alpha_5^{a_5} \beta_3^{b_3} \beta_4^{b_4} \beta_5^{b_5} \\ &= (p^3)^{a_3} (3p^3q)^{a_4} (6p^3q^2)^{a_5} (q^3)^{b_3} (3q^3p)^{b_4} (6q^3p^2)^{b_5} \\ &= Cp^{3a_3+3a_4+3a_5+b_4+2b_5} q^{3b_3+3b_4+3b_5+a_4+2a_5} \\ &= Cp^a q^b, \end{aligned}$$

where C is a constant independent of p , while a and b are the total numbers of sets won only by A and by B , respectively. We obtain

$$\log L = \log C + a \log p + b \log(1-p),$$

and the MLE of p can be easily found as

$$\hat{p} = \frac{a}{a+b} = \frac{\text{number of sets won by } A}{\text{number of sets played}}.$$

Observe that the number of matches, equal to $a_3 + a_4 + a_5 + b_3 + b_4 + b_5$, does not appear in the estimator of p . The fact that the last set in each match must be won by the winner of the match is not relevant here: the estimator is the same whether or not we fix in advance the number of sets (not matches!) to be played, and record the number of sets won by A .

Still another important property of maximum likelihood estimators is that they can be obtained from samples in which some data are only partially observable. To illustrate this property, we will find the MLE from a censored sample.

²³This assumption is important since men play “best out of five sets,” whereas women play “best out of three sets.”

■ **EXAMPLE 11.35**

Suppose that we want to estimate parameter λ in the $\text{EXP}(\lambda)$ distribution. For convenience of terminology, let us interpret the observations as lifetimes of some pieces of equipment. A typical experiment consists here of putting n pieces of the equipment to test and observing the lifetimes $X_1, X_2, \dots$. Suppose that the experiment is interrupted after some time T , and some tested items are still working. The data then have the form of a certain number of values $X_1, \dots, X_m$ of observed lifetimes, while about the remaining $n - m$ items we know only that $X_{m+1} > T, \dots, X_n > T$.

The likelihood of such a sample is obtained by multiplying the values of density function at points $X_1, \dots, X_m$ and the probabilities $P\{X_i > T\} = e^{-\lambda T}$ for $i = m + 1, \dots, n$. Therefore the likelihood is

$$L(\lambda) = \lambda e^{-\lambda X_1} \times \dots \times \lambda e^{-\lambda X_m} (e^{-\lambda T})^{n-m} = \lambda^m e^{-\lambda[X_1 + \dots + X_m + (n-m)T]}.$$

Since

$$\log L(\lambda) = m \log \lambda - \lambda[X_1 + \dots + X_m + (n-m)T],$$

the MLE of λ is easily found to be

$$\hat{\lambda} = \frac{m}{\sum_{i=1}^m X_i + (n-m)T}. \quad (11.54)$$

Finally, we will discuss consistency, asymptotic unbiasedness, and asymptotic efficiency of MLE's. We will not prove, or even formulate, the theorem, as it lies beyond the scope of this book. It is not difficult, however, to provide an informal explanation why MLE's, under some regularity assumptions, have the properties asserted.

The main idea is as follows: In Section 11.4 we introduced the function $J(X, \theta) = \frac{\partial}{\partial \theta} [\log f(X, \theta)]$, used to define the information $I(\theta)$ and the concept of efficient estimators. If $\mathbf{X}^{(n)} = (X_1, \dots, X_n)$ is a random vector from distribution $f(x, \theta)$, then $J(\mathbf{X}^{(n)}, \theta) = \sum_{i=1}^n J(X_i, \theta)$, with components $J(X_i, \theta)$ being iid random variables. We know from Section 11.4 that $E[J(X_i, \theta)] = 0$ and $\text{Var} J(X_i, \theta) = I(\theta)$. Consequently $E[J(\mathbf{X}^{(n)}, \theta)] = 0$ and $\text{Var}[J(\mathbf{X}^{(n)}, \theta)] = nI(\theta)$. By the central limit theorem of Lindeberg–Lévy (Theorem 9.6.1), we have

$$\frac{J(\mathbf{X}^{(n)}, \theta)}{\sqrt{nI(\theta)}} \xrightarrow{d} Z, \quad (11.55)$$

where Z is the standard normal random variable.

Let $T = T(\mathbf{X}^{(n)})$ be an unbiased efficient estimator of θ so that $E_\theta(T) = \theta$ and $\text{Var}_\theta(T) = 1/nI(\theta)$. We know also by (11.46) that

$$T = \gamma_1(\theta)J(\mathbf{X}^{(n)}, \theta) + \gamma_2(\theta),$$

so

$$E_\theta(T) = \gamma_1(\theta)E_\theta[J(\mathbf{X}^{(n)}, \theta)] + \gamma_2(\theta) = \gamma_2(\theta)$$

and

$$\text{Var}_\theta(T) = \gamma_1^2(\theta)E[J(\mathbf{X}^{(n)}, \theta)]^2 = \gamma_1^2(\theta)\text{Var}[J(\mathbf{X}^{(n)}, \theta)] = \gamma_1^2(\theta)nI(\theta).$$

Consequently, we must have

$$\gamma_2(\theta) = \theta, \quad \gamma_1^2(\theta) = \frac{1}{n^2 I^2(\theta)},$$

which implies that $\gamma_1(\theta) = \pm 1/nI(\theta)$. It follows now that

$$T = \pm \frac{1}{nI(\theta)} J(\mathbf{X}^{(n)}, \theta) + \theta,$$

which means that

$$\frac{J(\mathbf{X}^{(n)}, \theta)}{\sqrt{nI(\theta)}} = \pm \sqrt{nI(\theta)}(T - \theta). \quad (11.56)$$

The left-hand side of (11.56) has an asymptotic standard normal distribution by (11.55), and the same must be true for the right-hand side (under any choice of sign).

Consequently, we proved the following theorem:

Theorem 11.5.1 *If $T = T(X_1, \dots, X_n)$ is an efficient estimator of θ based on a random sample from the distribution $f(x, \theta)$, then the random variable*

$$\sqrt{nI(\theta)}(T - \theta)$$

asymptotically has the standard normal distribution.

Actually, the same theorem holds if efficient estimators T are replaced by maximum likelihood estimators $\hat{\theta}_n$ of θ , provided that these estimators are obtained by solving the equation

$$\frac{\partial L(\theta, \mathbf{X}^{(n)})}{\partial \theta} = J(\mathbf{X}^{(n)}, \theta) = 0$$

so that

$$J(\mathbf{X}^{(n)}, \hat{\theta}_n) = 0. \quad (11.57)$$

The proof requires imposing a number of technical assumptions, and is quite complicated, but the main idea is simple. We first expand the function $J(\mathbf{X}^{(n)}, \theta)$ about the point $\hat{\theta}_n$ so that

$$\begin{aligned} J(\mathbf{X}^{(n)}, \theta) &= J(\mathbf{X}^{(n)}, \hat{\theta}_n) + (\theta - \hat{\theta}_n) \left. \frac{\partial J}{\partial \theta} \right|_{\theta=\hat{\theta}_n} \\ &\quad + \frac{1}{2}(\theta - \hat{\theta}_n)^2 \left. \frac{\partial^2 J}{\partial \theta^2} \right|_{\theta=\hat{\theta}_n} + \dots \end{aligned}$$

We want to show that the second order term converges to 0 in probability so that it can be neglected in the limit, and the same is true for the sum of all higher order terms. Using (11.57), we then write

$$J(\mathbf{X}^{(n)}, \theta) \approx (\theta - \hat{\theta}_n) \left. \frac{\partial J}{\partial \theta} \right|_{\theta=\hat{\theta}_n}.$$

The left-hand side is now a sum of iid components, as in the proof of Theorem 11.5.1, and it obeys the central limit theorem. Finally, studying the behavior of the derivatives

$$\left. \frac{\partial J(\mathbf{X}^{(n)}, \theta)}{\partial \theta} \right|_{\theta=\hat{\theta}_n},$$

we establish that Theorem 11.5.1 holds also with T replaced by estimators $\hat{\theta}_n$ (with the same norming constants). This shows that $\hat{\theta}_n$ are asymptotically unbiased, asymptotically efficient, and have an asymptotic normal distribution with mean θ and variance $1/nI(\hat{\theta}_n)$. These properties make maximum likelihood estimators a preferred choice.

Least Squares Estimators

This method of estimation is dating back to Lagrange and Gauss—for an interesting account of its discovery see Stigler (1986). The basic setup is now different—the data are independent but they are not coming from the same distribution. For a given value u of some *independent* variable (random or not) U , observations (one or perhaps more) of some random variable Y_u are taken. It is assumed that

$$Y_u = Q(u) + \epsilon,$$

where $Q(u)$ is some function of u , and ϵ is a random variable (usually called *error*), such that $E(\epsilon) = 0$ and $\text{Var}(\epsilon) = \sigma^2 < \infty$. It follows that

$$E(Y_u) = Q(u) \quad \text{and} \quad \text{Var}(Y_u) = \sigma^2.$$

The function $Q(u)$ is usually called the *regression* of Y on u .

■ EXAMPLE 11.36

One of the common situations falling under the scheme above arises when we analyze some system that changes in time. Thus, u is the time of taking the observation of some attribute of the system, and $Q(u)$ is the expected value of the observed random variable Y_u , interpreted also as the “true” state of the system.

In some cases, we may take only one observation at any given time u ; in other cases, we may have a number of observations made at the same time.

■ EXAMPLE 11.37

In some cases variable U can be controlled by the experimenter. For example, a chemist may study the rate of a certain reaction in different temperatures. He then chooses the temperature level u_1 and observes the reaction rate one or more times. Then he changes the level to u_2 , repeats the observations, and so on. The numbers of observations need not be the same for different values u . In general, the choice of distinct values $u_1, \dots, u_k$ (as well as the choice of k) and the choice of numbers of observations $n_1, n_2, \dots, n_k$ to be made at selected points $u_1, \dots, u_k$, belongs to the *design* of the experiment.

Assume now that we have the experimental data for the design that can be described by the set of pairs

$$(u_i, n_i), \quad i = 1, \dots, k, \quad (11.58)$$

where u_i 's are distinct values of variable U and n_i 's are positive integers.

The data have form of the array $\{y_{ij}, i = 1, \dots, k, j = 1, \dots, n_i\}$, where y_{ij} is the value of random variable Y_{ij} , representing the j th observation for the value u_i of variable U . We assume that all Y_{ij} 's are independent, with

$$E(Y_{ij}) = Q(u_i) \quad \text{and} \quad \text{Var}(Y_{ij}) = \sigma^2. \quad (11.59)$$

In most typical cases, the functional form of Q is postulated and assumed to depend on some parameters. For instance, in the case of a linear regression model, we assume that

$$Q(u) = \beta u + \alpha, \quad (11.60)$$

where β and α are the slope and intercept of the regression line. In more complicated setups, we may postulate a quadratic regression

$$Q(u) = \beta_2 u^2 + \beta_1 u + \alpha,$$

or some other functional form of Q . Generally,

$$Q(u) = \varphi(u; \theta_1, \theta_2, \dots, \theta_r),$$

where φ is a known function and $\theta_1, \dots, \theta_r$ are some parameters to be estimated. The method of *least squares* is based on the quadratic form

$$S(\theta_1, \dots, \theta_r) = \sum_{i=1}^k \sum_{j=1}^{n_i} [y_{ij} - \varphi(u_i; \theta_1, \dots, \theta_r)]^2. \quad (11.61)$$

The values $\hat{\theta}_1, \dots, \hat{\theta}_r$ that minimize S given by (11.61) are called least squares (LS) estimates of $\theta_1, \dots, \theta_r$. As usual, those estimators, regarded as functions of the random variables $\{Y_{ij}\}$ are called *LS-estimators*.

The usual way of finding $\hat{\theta}_1, \dots, \hat{\theta}_r$ is by solving the set of simultaneous equations

$$\frac{\partial S}{\partial \theta_l} = 0, \quad l = 1, \dots, r, \quad (11.62)$$

which in the present case take the form

$$\sum_{j=1}^{n_i} [y_{ij} - \varphi(u_i; \theta_1, \dots, \theta_r)] \sum_{i=1}^k \frac{\partial \varphi(u_i; \theta_1, \dots, \theta_r)}{\partial \theta_l} = 0. \quad (11.63)$$

■ EXAMPLE 11.38 Linear Regression

Suppose that $Q(u) = \beta u + \alpha$. In this case the algebra will simplify somewhat if we order all Y_{ij} 's into a single sequence $y_1, \dots, y_n$, where $n = \sum_{i=1}^k n_i$, and relabel the corresponding values as $u_1, \dots, u_n$ (n values u_i of variable U , not necessarily all distinct). The quantity to be minimized is

$$S(\alpha, \beta) = \sum_{i=1}^n (y_i - \beta u_i - \alpha)^2. \quad (11.64)$$

Differentiating with respect to α and β and setting the derivatives equal to 0, we obtain

$$\beta \sum_{i=1}^n u_i^2 + \alpha \sum_{i=1}^n u_i = \sum_{i=1}^n u_i y_i \quad \text{and} \quad \beta \sum_{i=1}^n u_i + n\alpha = \sum_{i=1}^n y_i.$$

The solution can be written as

$$\hat{\beta} = \frac{\sum (u_i y_i - \bar{u} \times \bar{y})}{\sum (u_i - \bar{u})^2}, \quad (11.65)$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{u}, \quad (11.66)$$

provided that $\sum (u_i - \bar{u})^2 > 0$. The latter condition is ensured if there are at least two distinct values of u used in the experiment, that is, if not all observations are made for the same value of S .

An easy check shows that (11.65) does minimize function U .

Robust Estimators

An important issue in estimation problems is to obtain estimators, generally termed *robust*, that would be relatively unaffected by deviations from the assumed model. We will now briefly sketch two such approaches.

To grasp the main issues involved here, consider the problem of estimating the mean μ of a distribution. Then, the sample mean $\bar{X}$ is an estimator of μ , and we know that $\bar{X}$ has a number of desirable properties, such as consistency, unbiasedness, and efficiency. These properties, however, are valid under specific assumptions about the underlying distribution. If the actual distribution differs in some way from the assumed one, $\bar{X}$ may no longer have the same properties. We say that $\bar{X}$ is sensitive to the deviation from the model, or *contamination*. This concept is intended to describe the occurrence of *outliers*, that is, observations that follow a distribution different from the one assumed in the model. Most of the outlier distribution is usually concentrated around much larger (or smaller) values than those typically encountered in the model. Formally, the model assumes the distribution $f(x, \theta)$, where θ is (say) the mean, while the actual sample is taken from the distribution

$$\varphi(x, \theta) = (1 - \epsilon)f(x, \theta) + \epsilon g(x), \quad (11.67)$$

where most of the mass of $g(x)$ is far away from the range of “typical” values under $f(x, \theta)$.

The two important approaches to robust estimation are L -estimators and M -estimators.

L -Estimators L -estimators, linear functions of order statistics, provide good estimators of location and scale parameters.

Formally, θ will be called a *location parameter* if

$$f(x, \theta) = h(x - \theta)$$

for some probability density (or probability mass function) h .

Similarly, θ is called a *scale parameter* if

$$f(x, \theta) = \frac{1}{\theta} h\left(\frac{x}{\theta}\right)$$

for some probability density (or probability mass function) h .

The mean of a distribution may or may not be its location parameter. For instance, it is so in the case of normal distribution. However, in the case of exponential distribution, the mean is a scale parameter, while in the case of a gamma distribution, the mean is neither a scale or a location parameter.

Let $X_1, \dots, X_n$ be a random sample from the distribution $f(x, \theta)$, and let

$$X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$$

denote the order statistics of the sample. The L -estimator will be any statistic T of the form

$$T = \sum_{k=1}^n \gamma_{n,k} X_{k:n},$$

where $\gamma_{n,k}$ for $k = 1, \dots, n$ is a double array of coefficients. The class of L -estimators contains many well-known estimators: Choosing $\gamma_{n,k} = 1/n$ for $k = 1, \dots, n$ gives $T = \bar{X}$. The choice $\gamma_{n,1} = 1, \gamma_{n,k} = 0$ for $k \geq 2$, or $\gamma_{n,n} = 1, \gamma_{n,k} = 0$ for $k < n$, gives two extreme order statistics $X_{1:n}$ and $X_{n:n}$. In a similar way, one can obtain any sample quantile. Choosing $\gamma_{n,[3n/4]+1} = 1, \gamma_{n,[n/4]+1} = -1, \gamma_{n,k} = 0$ for remaining k , one obtains a sample interquartile range, and so on.

Perhaps the most important L -estimators are the trimmed and Winsorized means, defined as follows:

Definition 11.5.2 Let $0 < \alpha < 1/2$. Then, the α -trimmed mean is

$$U_n = \frac{1}{n - 2[n\alpha]} \sum_{k=[n\alpha]+1}^{n-[n\alpha]} X_{k:n}, \quad (11.68)$$

while the α -Winsorized mean is

$$V_n = \frac{1}{n} \left([n\alpha] X_{[n\alpha]+1:n} + \sum_{k=[n\alpha]+1}^{n-[n\alpha]} X_{k:n} + [n\alpha] X_{n-[n\alpha]:n} \right). \quad (11.69)$$

□

On the one hand, α -trimming consists of removing the $100\alpha\%$ of lowest and highest observations from the sample, and taking an average of the remaining ones (the middle $100(1 - 2\alpha)\%$ of observations). On the other hand, α -Winsorizing consists of replacing each observation in the lower $100\alpha\%$ and in the upper $100\alpha\%$ of the sample by the sample quantile of order α and $1 - \alpha$, respectively, so that the sample size will stay unchanged. The Winsorized mean is then calculated as the mean of the Winsorized sample.

It is clear that the purpose of trimming (or Winsorizing) is to eliminate (or decrease) the effect of outliers in the sample. An important issue is to define the notion of optimality and then to determine the optimal level α at which the mean should be trimmed or Winsorized.

Among the main advantages of L -estimators is their asymptotic normality when the weights $\gamma_{n,k}$ are reasonably smooth or if they are nonzero for a certain number of central order statistics. For more details, see Serfling (1980) and Arnold et al. (1992).

M -Estimators Another class of estimators is obtained as follows: Let $h(x, u)$ be a function of two arguments. Given a sample $x_1, x_2, \dots, x_n$ from the distribution $f(x, \theta)$, take as an estimator of θ the solution of the equation

$$\sum_{k=1}^n h(x_k, u) = 0. \quad (11.70)$$

Such estimators are most often obtained by solving an approximate minimization problem. Suppose that we have a “distance” of some sort (not necessarily satisfying any conditions for metric), say $H(x, u)$. As an estimator of θ , we choose a point u^* that minimizes the sum

$$\sum_{k=1}^n H(x_k, u), \quad (11.71)$$

interpreted as the sum of distances from u to all sample points. In a sense, u^* is the point closest to the sample, with closeness being expressed by the function H . Differentiating (11.71) with respect to u and setting the derivative equal to 0, we obtain equation (11.70) with $h(x, u) = \frac{\partial}{\partial u} H(x, u)$.

This formulation comprises two important classes of estimators, namely MLE's and LS-estimators. Indeed, if we define the function $H(x, u)$ as $-\log f(x; u)$, then $h(x, u) = -\frac{\partial}{\partial u} \log f(x, u)$ and the M -estimator corresponding to this choice is the maximum likelihood estimator (the minus sign is implied by the fact that now the problem is formulated as a minimization problem).

By taking appropriate functions H and h , we can obtain different variants of least square estimators. Similarly, trimmed or Winsorized means can be obtained by appropriate choices of the functions H and h . For example, we may take $H(x, u) = H(x - u)$ for some function H of one argument. The M -estimator then minimizes the sum $\sum_{i=1}^n H(X_i - u)$. For $H(x) =$

x^2 , we have the simplest least square estimator. If $H(x) = x^2$ for $|x| \leq k$ and $H(x) = k^2$ for $|x| > k$, we obtain a form of the Winsorized mean.

As with L -estimators, the main direction of research is to study the asymptotic properties (e.g., normality) of M -estimators under some general assumptions on functions H or h , and distributions of X_i .

PROBLEMS

11.5.1 Let $X_1, \dots, X_n$ be a random sample from $\text{GAM}(\alpha, \lambda)$ distribution. Find: **(i)** The MME of $\theta = (\alpha, \lambda)$, using the first two moments. **(ii)** The MME of α when λ is known, and the MME of λ when α is known.

11.5.2 Find the MME of parameter θ in the distribution with density $f(x, \theta) = (\theta + 1)x^{-(\theta+2)}$, for $x > 1, \theta > 0$.

11.5.3 Let $X_1, \dots, X_n$ be a random sample from the distribution uniform on the union of the two intervals: $[-2, -1]$ and $[0, \theta]$. Find: **(i)** The MME of θ . **(ii)** The MLE of θ . **(iii)** The MLE of θ if positive X_i 's are recorded exactly, and negative X_i 's can only be counted. **(iv)** The MLE of θ if X_i 's cannot be observed, and one can only count the numbers of positive and negative ones.

11.5.4 Let $X_1, \dots, X_n$ be a random sample from Poisson distribution with mean λ . Find the MLE of $P(X = 0)$.

11.5.5 (Brugging Tennis Player) As in Example 11.34, consider tennis players A and B who from time to time play matches against each other. The probability that A wins a set against B is p .

Suppose now that we do not have complete data on all matches between A and B ; we learn only of A 's victories, so we know numbers a_3, a_4 , and a_5 of matches won in 3, 4, and 5 sets (we do not even know whether he lost any matches with B).

Show that one can find the MLE of p , and find the MLE of the total number of matches and of the number of sets that A lost against B (do not attempt the algebraic solution).

11.5.6 Some phenomena (e.g., headway in traffic) are modeled to have a distribution of a sum of a constant and an exponential random variable. Then the density of X has the form

$$f(x; a, b) = \begin{cases} 0 & \text{for } x < b \\ ae^{-a(x-b)} & \text{for } x \geq b, \end{cases}$$

where $a > 0$ and $b > 0$ are two parameters. Find: **(i)** The MME of $\theta = (a, b)$. **(ii)** The MLE of θ .

11.5.7 A single observation of a random variable X with a geometric distribution results in $X = k$. Find the MLE of the probability of success θ if: **(i)** X is the number of failures preceding the first success. **(ii)** X is the number of trials up to and including the first success.

11.5.8 Find the distribution of the MLE of the probability of success θ based on two Bernoulli trials.

11.5.9 Let $X_1, \dots, X_n$ be a random sample from $N(\theta, \sigma^2)$ distribution, $\theta \neq 0, \sigma^2$ is known. Determine the asymptotic distribution of $(\bar{X})^{1/2}$ using the invariance property of MLEs.

- 11.5.10** Let $X_1, \dots, X_n$ be a random sample from a distribution with density $f(x, \theta) = 3\theta^{-3}x^2$ for $0 \leq x \leq \theta$. **(i)** Find the MLE of θ . **(ii)** Find the MME of θ . **(iii)** Approximate the distribution of $T = 3\bar{X}$. **(iv)** Obtain the MSE of $T_a = a\bar{X}$ as an estimator of θ .
- 11.5.11** Suppose that there were 15 successes in 24 Bernoulli trials. Find the MLE of the probability of success θ if it is known that $\theta \leq 1/2$.
- 11.5.12** Two independent Bernoulli trials resulted in one failure and one success. What is the MLE of the probability of success θ if it is known that: **(i)** θ is at most $1/4$. **(ii)** θ exceeds $1/4$.
- 11.5.13** Let $X_1, \dots, X_n$ be a random sample from $\text{POI}(\lambda)$ distribution. Find the MLE of λ assuming that: **(i)** $X_1 + \dots + X_n > 0$. **(ii)** $X_1 + \dots + X_n = 0$.
- 11.5.14** Find the MME and MLE of the standard deviation of a Poisson distribution.
- 11.5.15** Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ distribution, μ is known. Find the MLE of σ : **(i)** Directly. **(ii)** First finding the MLE of variance σ^2 and then using the invariance property.
- 11.5.16** Let X_1, X_2 be a random sample from a $N(\mu, \sigma^2)$ distribution with μ and σ^2 unknown. Find the MLE of σ^2 if the only available information is that the difference between observations equals 3.
- 11.5.17** Find the MLE of the mean of a $U[\theta_1, \theta_2]$ distribution based on a random sample of size n .
- 11.5.18** For n observations taken from the $U[0, \theta]$ distribution, let U_n be the number of the ones that are less than 3. Find the MLE of θ .
- 11.5.19** Suppose that the median of 20 observations, taken from a normal distribution with an unknown mean and variance, is 5 and that only one observation differs from the median by more than 3. Suggest an estimate of the probability that the next two observations will both be between 4 and 5.
- 11.5.20** Let $X_1, \dots, X_n$ be a random sample from a log-normal distribution with parameters μ and σ^2 [this means that $\log X_i \sim N(\mu, \sigma^2)$]. Find the MLE of μ and σ^2 .
- 11.5.21** Let $X_1, \dots, X_m$ be a random sample from the $N(\mu_1, \sigma_1^2)$ distribution and let $Y_1, \dots, Y_n$ be a random sample from the $N(\mu_2, \sigma_2^2)$ distribution, with X_i 's being independent from Y_j 's. Find the MLE of: **(i)** μ_1, μ_2, σ^2 if $\sigma_1 = \sigma_2 = \sigma$. **(ii)** $\mu, \sigma_1^2, \sigma_2^2$ where $\mu_1 = \mu_2 = \mu$.
- 11.5.22** $Y_1, \dots, Y_n$ are independent variables. Assuming that $x_1, \dots, x_n$ are such that $\sum (x_i - \bar{x})^2 > 0$, compare the MSE's of the MLE and LS-estimators of parameter $\theta > 0$, if: **(i)** $Y_i \sim \text{EXP}(x_i/\theta)$. **(ii)** $Y_i \sim \text{POI}(\theta x_i)$.
- 11.5.23** For independent variables $Y_1, \dots, Y_n$ with distribution $N(\alpha + \beta x_i, \sigma^2)$, where $x_1, \dots, x_n$ are fixed, show that the LS-estimator and ML-estimator of $\theta = (\alpha, \beta)$ coincide.
- 11.5.24** Let $X_1, \dots, X_n$ be a random sample from $U[\theta, \theta + 1]$ distribution. **(i)** Show that $T = c(X_{n:n} - 1) + (1 - c)X_{1:n}$, $0 < c < 1$, is the MLE of θ , and find the value of c that minimizes its MSE. **(ii)** Determine the asymptotic distribution of T if $c = 0.5$.
- 11.5.25** Let $X_1, \dots, X_n$ be a random sample from a distribution with density $f(x, \theta) = \theta(1 + x)^{-(1+\theta)}$ and for $x > 0$ and 0 otherwise, $\theta \geq 1$. Find the MME of θ .

- 11.5.26** Let $X_1, \dots, X_n$ be independent observations from a distribution with density $f(x, \theta) = 2x/\theta^2$, for $0 \leq x \leq \theta$ and 0 otherwise. Find the MME of θ and approximate its distribution.
- 11.5.27** Let $X_1, \dots, X_n$ be a random sample from $\text{EXP}(\theta)$ distribution. Find the MLE of the second factorial moment $E(X(X-1))$ and determine its asymptotic distribution.

11.6 SUFFICIENCY

The considerations of this section are motivated by the following observation: each of the estimators analyzed in this chapter is dependent on the random sample $X_1, \dots, X_n$ through some statistic (e.g., $\bar{X}$ or $X_{n:n}$), that reduces the data to a single number. From a purely formal viewpoint, any such reduction involves some loss of information.

The concept of a sufficient statistic is intended to cover situations where the information lost in reducing data to the value of a statistic is not relevant for the purpose of estimating parameter θ . Thus the definition of sufficiency of a statistic is relative to a given parameter θ .

As before, we consider a random sample $X_1, \dots, X_n$ from the distribution $f(x, \theta)$, where θ belongs to some parameter space Θ . Again, f can be either a density or a probability function of a discrete distribution.

The definition below introduces a statistic that conveys the same information about θ as the whole sample $(X_1, \dots, X_n)$.

Definition 11.6.1 The statistic T is said to be *sufficient* for θ if the conditional distribution of $X_1, \dots, X_n$, given $T = t$, does not depend on θ for any value t . $\square$

Before exploring the consequences of the definition let us analyze some examples.

■ EXAMPLE 11.39

Consider two independent Bernoulli trials X_1, X_2 , with probability of success p , and let $T = X_1 + X_2$. The conditional distribution of (X_1, X_2) given T is

$$P\{X_1 = 0, X_2 = 0 | T = 0\} = 1,$$

$$P\{X_1 = 1, X_2 = 1 | T = 2\} = 1.$$

Since $P(T = 1) = P(X_1 = 1, X_2 = 0) + P(X_1 = 0, X_2 = 1) = 2p(1-p)$, for $T = 1$ we have

$$P\{X_1 = 1, X_2 = 0 | T = 1\} = \frac{P\{X_1 = 1, X_2 = 0\}}{P(T = 1)} = \frac{p(1-p)}{2p(1-p)} = \frac{1}{2}.$$

Thus, whatever the value of T (0, 1, or 2), the conditional distribution of (X_1, X_2) does not depend on p .

■ EXAMPLE 11.40

Let X_t and X_s be the numbers of events in a Poisson process with intensity λ , observed between 0 and t , and between t and $t + s$. We will show that $U = X_t + X_s$ is a sufficient statistic for λ .

Indeed, since X_t and X_s are independent, we have for $k = 0, 1, \dots, n$,

$$\begin{aligned} P\{X_t = k, X_s = n - k | U = n\} &= \frac{P\{X_t = k\}P\{X_s = n - k\}}{P\{U = n\}} \\ &= \frac{\frac{(\lambda t)^k}{k!} e^{-\lambda t} \times \frac{(\lambda s)^{n-k}}{(n-k)!} e^{-\lambda s}}{\frac{[\lambda(t+s)]^n}{n!} e^{-\lambda(t+s)}} \\ &= \binom{n}{k} \left(\frac{t}{t+s}\right)^k \left(\frac{s}{t+s}\right)^{n-k}. \end{aligned}$$

Thus X_t , given $U = n$, is binomial with parameters n and $t/(t+s)$, and does not depend on λ .

If the conditional distribution of observations $X_1, \dots, X_n$, given the statistic $T = t$, does not depend on the parameter θ that we want to estimate, then (once we know that $T = t$) the additional knowledge of a particular configuration of $X_1, \dots, X_n$ observed in the data is irrelevant in estimating θ . For instance, in Example 11.39, $T = 1$ means that we had one success and one failure. The fact that the first trial was a failure and the second was a success is of no additional help in estimating p .

Since the process of determining the conditional distribution of observations given the value of statistic T is sometimes cumbersome, it is desirable to have another method for verifying that a statistic is sufficient. Such a method is given by the following theorem, due to Neyman:

Theorem 11.6.1 (Factorization Criterion) *Let $X_1, \dots, X_n$ be a random sample from the distribution $f(x, \theta)$, with $\theta \in \Theta$. A statistic $T = T(X_1, \dots, X_n)$ is a sufficient statistic for θ if and only if for all $\mathbf{x} = (x_1, \dots, x_n)$ and all $\theta \in \Theta$ the joint distribution*

$$f_n(\mathbf{x}, \theta) = f(x_1, \theta) \cdots f(x_n, \theta)$$

can be written as

$$f_n(\mathbf{x}, \theta) = u[T(\mathbf{x}), \theta]v(\mathbf{x}), \quad (11.72)$$

where u is a nonnegative function that depends on both θ and $\mathbf{x} = (x_1, \dots, x_n)$, but dependence on $\mathbf{x}$ is only through the function T , and v is a function of $\mathbf{x} = (x_1, \dots, x_n)$ that does not depend on θ .

Proof: We will give the proof in the discrete case; the proof for the continuous case requires careful consideration because densities are defined only up to sets of probability zero. Thus, we have now $f(x, \theta) = P_\theta\{X_i = x\}$, $i = 1, \dots, n$.

Let $Q(t)$ be the set of all $\mathbf{x} = (x_1, \dots, x_n)$ such that $T(\mathbf{x}) = t$. Then

$$P_\theta(T = t) = \sum_{\mathbf{x} \in Q(t)} f_n(\mathbf{x}, \theta).$$

Suppose first that T is a sufficient statistic. Then for $T = t$, and any point $\mathbf{x} \in Q(t)$ the conditional probability $P_\theta\{\mathbf{X} = \mathbf{x} | T = t\}$ does not depend on θ ; let us call it $v(\mathbf{x})$. Letting $u(t, \theta) = P_\theta\{T = t\}$, we obtain, for any fixed $\mathbf{x} \in Q(t)$,

$$f_n(\mathbf{x}, \theta) = P_\theta\{\mathbf{X} = \mathbf{x}\} = P_\theta\{\mathbf{X} = \mathbf{x} | T = t\}P_\theta\{T = t\} = v(\mathbf{x})u(t, \theta),$$

which is the factorization (11.72).

Conversely, suppose that $f_n(\mathbf{x}, \theta)$ satisfies formula (11.72) for some functions u and v . Let us fix t and $\theta \in \Theta$ and compute the conditional probability of a point $\mathbf{x}$ given $T = t$. Clearly, for $\mathbf{x} \notin Q(t)$, we have $P_\theta\{\mathbf{X} = \mathbf{x} | T = t\} = 0$. For $\mathbf{x} \in Q(t)$, we have

$$\begin{aligned} P_\theta\{\mathbf{X} = \mathbf{x} | T = t\} &= \frac{P_\theta\{\mathbf{X} = \mathbf{x}\}}{P_\theta\{T = t\}} = \frac{f_n(\mathbf{x}, \theta)}{\sum_{\mathbf{y} \in Q(t)} f_n(\mathbf{y}, \theta)} \\ &= \frac{u(t, \theta)v(\mathbf{x})}{\sum_{\mathbf{y} \in Q(t)} u(t, \theta)v(\mathbf{y})} = \frac{v(\mathbf{x})}{\sum_{\mathbf{y} \in Q(t)} v(\mathbf{y})}. \end{aligned}$$

□

We will now apply Theorem 11.6.1 to find sufficient statistics in various families of distributions.

■ EXAMPLE 11.41

Consider again the Bernoulli trials with probability p of success. We have here, letting $t = \sum x_i$,

$$\begin{aligned} f_n(x_1, \dots, x_n, p) &= \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^t (1-p)^{n-t} \\ &= \left(\frac{p}{1-p} \right)^t (1-p)^n. \end{aligned}$$

In this case, a sufficient statistic is the total number of successes $T = \sum X_i$, while the function $v(x_1, \dots, x_n) \equiv 1$.

■ EXAMPLE 11.42

In a Poisson distribution, we have, letting $t = \sum x_i$,

$$f_n(x_1, \dots, x_n, \lambda) = \prod_{i=1}^n \frac{\lambda^{x_i}}{x_i!} e^{-\lambda} = \lambda^t e^{-n\lambda} \frac{1}{x_1! \cdots x_n!},$$

which again shows that $T = \sum X_i$ is a sufficient statistic. Here $u(t, \lambda) = \lambda^t e^{-n\lambda}$, while $v(x_1, \dots, x_n) = (x_1! \cdots x_n!)^{-1}$.

■ EXAMPLE 11.43

Now, let $(x_1, \dots, x_n)$ be observations from a normal distribution $N(\mu, \sigma^2)$ where σ^2 is known. Then, the joint density of the sample can be written as

$$\begin{aligned} f_n(x_1, \dots, x_n, \mu) &= \frac{1}{\sigma^n (2\pi)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right\} \\ &= \frac{1}{\sigma^n (2\pi)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \left(\sum x_i^2 - 2\mu \sum x_i + n\mu^2 \right) \right\} \\ &= \exp \left\{ \frac{\mu}{\sigma^2} \sum x_i - \frac{n\mu^2}{2\sigma^2} \right\} v(x_1, \dots, x_n). \end{aligned}$$

Thus, $\sum X_i$ is a sufficient statistic for μ , with

$$v(x_1, \dots, x_n) = \frac{\exp \{ -1/(2\sigma^2) \sum x_i^2 \}}{\sigma^n (2\pi)^{n/2}}.$$

If now μ is known, but σ^2 is the unknown parameter, we take $v(x_1, \dots, x_n) \equiv 1$, and the sufficient statistic is $\sum (X_i - \mu)^2$.

■ EXAMPLE 11.44

Consider finally the case of sampling from a $U[0, \theta]$ distribution. The joint density of the sample is

$$f(x_1, \dots, x_n, \theta) = \begin{cases} \frac{1}{\theta^n} & \text{if } X_{n:n} \leq \theta \\ 0 & \text{otherwise,} \end{cases}$$

or equivalently

$$f(x_1, \dots, x_n, \theta) = \frac{1}{\theta^n} I(\theta, \max(x_1, \dots, x_n)),$$

where $I(\theta, c) = 1$ if $\theta \geq c$ and 0 if $\theta < c$. This shows the function $u(\theta, T)$ (with $v(x_1, \dots, x_n) \equiv 1$) and that $X_{n:n}$ is a sufficient statistic.

Sufficient statistics are not unique. Every one-to-one transformation of a sufficient statistic is again a sufficient statistic. For instance, in Example 11.43, an alternative sufficient statistic to $\sum X_i$ is $\bar{X} = (1/n) \sum X_i$.

The concept of sufficient statistic can be generalized in an obvious way to a set of jointly sufficient statistics to cover a multidimensional case. The intuitions motivating the concept are the same as in a one-dimensional case.

We will now formulate the definition using the factorization theorem, that is, use the assertion of the theorem as a basis for the definition.

Definition 11.6.2 The statistics $T_1, \dots, T_k$ are minimal *jointly sufficient* for θ if the distribution $f_n(\mathbf{x}, \theta)$ satisfies the following condition: for every $\mathbf{x} = (x_1, \dots, x_n)$ and $\theta \in \Theta$ we have

$$f_n(\mathbf{x}, \theta) = u[T_1(\mathbf{x}), \dots, T_k(\mathbf{x}), \theta] v(\mathbf{x}),$$

where the nonnegative function $v(\mathbf{x})$ does not depend on θ while the function u depends on θ , and depends on $\mathbf{x}$ only through the values of statistics $T_1, \dots, T_k$. $\square$

■ EXAMPLE 11.45

In a random sample $X_1, \dots, X_k$ from a $N(\mu, \sigma^2)$ distribution, where both μ and σ^2 are unknown so that $\theta = (\mu, \sigma^2)$, we have

$$\begin{aligned} f_n(\mathbf{x}, \theta) &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right\} \\ &= \frac{\sigma^{-n}}{(2\pi)^{n/2}} \exp \left\{ -\frac{\sum x_i^2}{2\sigma^2} + \frac{\mu}{\sigma^2} \sum x_i - \frac{n\mu^2}{2\sigma^2} \right\}; \end{aligned}$$

statistics $T_1 = \sum_{i=1}^n X_i$ and $T_2 = \sum_{i=1}^n X_i^2$ are jointly sufficient for θ .

If $T_1, \dots, T_k$ are jointly sufficient for θ , and $T_1^*, \dots, T_k^*$ are obtained from $T_1, \dots, T_k$ by a one-to-one transformation, then $T_1^*, \dots, T_k^*$ are also jointly sufficient for θ . Thus, another

pair of jointly sufficient statistics in Example 11.45 is $\bar{X}$ and $S^2 = (1/n) \sum_{i=1}^n (X_i - \bar{X})^2$. This can also be seen from the representation

$$\begin{aligned} f_n(\mathbf{x}, \theta) &= (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right\} \\ &= (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \bar{X})^2 - \frac{n(\bar{X} - \mu)^2}{2\sigma^2} \right\} \\ &= (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp \left\{ -\frac{n}{2\sigma^2} [S^2 - (\bar{X} - \mu)^2] \right\}. \end{aligned}$$

Obviously, the sample $(X_1, \dots, X_n)$ is always jointly sufficient. It suffices to write formally $T_i(X_1, \dots, X_n) = X_i$ for $i = 0, 1, \dots, n$. This is true but useless. It is also true that the order statistics $(X_{1:n}, X_{2:n}, \dots, X_{n:n})$ are jointly sufficient, since

$$f_n(\mathbf{x}, \theta) = \prod_{j=1}^n f(x_j, \theta) = \prod_{i=1}^n f(x_{i:n}, \theta),$$

where the factors of the last product are a permutation of the factors of the middle product.

Thus, a set of jointly sufficient statistics always exists, and it is natural to look for the maximal reduction of data that retains sufficiency. Here reduction is meant both in the sense of reducing the dimensionality, and in the sense of using transformations that are not one-to-one, such as squaring.

The definition of maximal reduction, hence minimality of sufficient statistic, is as follows:

Definition 11.6.3 A sufficient statistic T is called *minimal* if T is a function of any other sufficient statistic. $\square$

The definition covers the idea of maximal reduction of the data, still leaving sufficiency intact. Indeed, suppose that T is minimal in the sense of Definition 11.6.3 and still can be reduced. This means that there exists a function h which is not one-to-one such that $U = h(T)$ is a sufficient statistic. Then, T is not a function of U , which gives a contradiction.

Definition 11.6.3 extends to the case of a minimal sufficient set of statistics.

Definition 11.6.4 A set $\{T_1, \dots, T_k\}$ of sufficient statistics is called *minimal* if $T_1, \dots, T_k$ are jointly sufficient and they are functions of any other set of jointly sufficient statistics. $\square$

The intuition here is as follows: if $\{T_1, \dots, T_m\}$ and $\{T'_1, \dots, T'_k\}$ are both jointly sufficient, and the second set is a function of the first, that is, if $T'_i = \varphi_i(T_1, \dots, T_m)$ for $i = 1, \dots, k$ and for some functions $\varphi_1, \dots, \varphi_k$, then

$$(T_1, \dots, T_m) \succeq (T'_1, \dots, T'_k),$$

where $\succeq$ is an ordering among jointly sufficient sets of statistics. The minimal jointly sufficient statistics are at the “base” of this ordering, while the fact that $(X_{1:n}, \dots, X_{n:n})$ is always a set of jointly sufficient statistics means that the set of statistics being ordered by relation $\succeq$ is not empty. Example 11.45 shows that the minimal sufficient set of statistics is not unique; however, all these minimal sufficient sets are obtained as one-to-one functions of other minimal sufficient sets.

■ **EXAMPLE 11.46**

Let $X_1, \dots, X_n$ be a random sample from the distribution uniform on $[-\theta, \theta]$ so that

$$f(x; \theta) = \begin{cases} \frac{1}{2\theta} & \text{if } -\theta \leq x \leq \theta \\ 0 & \text{otherwise.} \end{cases}$$

Proceeding as in Example 11.44, we can easily show that the pair of extreme order statistics $(X_{1:n}, X_{n:n})$ is jointly sufficient for θ . However, this is not a minimal set: the statistic $T(X_{1:n}, X_{n:n}) = \max(|X_{1:n}|, |X_{n:n}|)$ is a single sufficient statistic for θ .

Typically, the dimensionality of the minimal set of jointly sufficient statistics equals the number of dimensions of the parameter θ . Thus, in the case of a normal distribution with the parameter being the pair (μ, σ^2) , the minimal sets of jointly sufficient statistics consist of two statistics (e.g., $\sum X_i$ and $\sum X_i^2$). However, there exist distributions that depend on a single parameter, and yet the only jointly sufficient statistics (hence the minimal sets of jointly sufficient statistics) are the order statistics $(X_{1:n}, \dots, X_{n:n})$. This is true, for instance, for the Cauchy distribution with density

$$f(x; \theta) = \frac{1}{\pi[1 + (x - \theta)^2]}.$$

Consider now the role that sufficient statistics play in estimation. Observe first that the likelihood function is

$$L(\theta, \mathbf{x}) = v(\mathbf{x})u[T_1(\mathbf{x}), \dots, T_k(\mathbf{x}), \theta], \quad (11.73)$$

where $\{T_1, \dots, T_k\}$ is a set of jointly sufficient statistics (minimal or not). Since the factor $v(\mathbf{x})$ plays no role in the determination of θ at which the likelihood attains its maximum, we can expect that the MLE $\hat{\theta}$ of θ will be a function of $T_1, \dots, T_k$. Since the same kind of representation of the likelihood is valid if $T_1, \dots, T_k$ form a minimal sufficient set, we can expect that $\hat{\theta}$ will be a function of a minimal sufficient set of statistics.

This argument is valid provided that the MLE is unique (which is usually the case). When the MLE is not unique (for such situations, see Example 11.30), one cannot claim that they are all functions of sufficient statistics, but one of them will be.

We can therefore state the following theorem:

Theorem 11.6.2 *If the MLE of θ exists and is unique, then it is a function of minimal sufficient set of statistics. If the MLE is not unique, then there exists an MLE that is a function of a minimal sufficient set of statistics.*

The main importance of sufficient statistics is, to a large extent, related to the next theorem, according to which any estimator that is not based on sufficient statistic can be improved, and hence is not admissible (see Definition 11.3.1). The theorem is formulated in the case of a single sufficient statistic T . However, it can be easily generalized to the case of the minimal sufficient set of statistics.

Theorem 11.6.3 Rao–Blackwell *Let T be a statistic sufficient for θ in a family of distributions $\{f(x, \theta)\}$ and let W be an estimator of $g(\theta)$. Statistic*

$$W^*(T) = E_\theta(W|T) \quad (11.74)$$

is also an estimator of $g(\theta)$. Moreover, $E(W^) = E(W)$ and its risk $R(W^*, \theta) = E\{[W^* - g(\theta)]^2\}$ satisfies the condition*

$$R(W^*, \theta) \leq R(W, \theta) \quad (11.75)$$

for all $\theta \in \Theta$. The inequality in (11.75) is strict for some θ unless W is a function of T and is an unbiased estimator of $g(\theta)$.

Proof: We need to show first that formula (11.74) indeed defines an estimator of $g(\theta)$, that is, that the left-hand side does not depend on θ but only on the sample $X_1, \dots, X_n$. We will also show that this dependence is through the value of T only so that the obtained estimator is a function of the sufficient statistic T . The second part of the proof will consist of showing that inequality (11.75) holds for every θ .

To show that formula (11.74) defines an estimator, we will carry the argument in the discrete case, with $f(x, \theta) = P\{X = x|\theta\}$. Let t be a possible value of T , and let also $Q(t) = \{\mathbf{x} : T(\mathbf{x}) = t\}$.

For any fixed t and $\mathbf{x} \in Q(t)$, the conditional distribution of $\mathbf{X}$ given $T = t$ is, by the factorization theorem,

$$\begin{aligned} P\{\mathbf{X} = \mathbf{x} | T = t, \theta\} &= \frac{f_n(\mathbf{x}, \theta)}{\sum_{\mathbf{y} \in Q(t)} f_n(\mathbf{y}, \theta)} = \frac{v(\mathbf{x})u(t, \theta)}{\sum_{\mathbf{y} \in Q(t)} v(\mathbf{y})u(t, \theta)} \\ &= \frac{v(\mathbf{x})}{\sum_{\mathbf{y} \in Q(t)} v(\mathbf{y})}, \end{aligned}$$

which is independent of θ . On the other hand, $P\{\mathbf{X} = \mathbf{x} | T = t, \theta\} = 0$ if $\mathbf{x} \notin Q(t)$, which is also independent of θ . Consequently, we have

$$E_\theta(V|T = t) = \sum_{\mathbf{x} \in Q(t)} W(\mathbf{x})P\{\mathbf{X} = \mathbf{x} | T = t, \theta\} = \frac{\sum_{\mathbf{x} \in Q(t)} W(\mathbf{x})v(\mathbf{x})}{\sum_{\mathbf{y} \in Q(t)} v(\mathbf{y})},$$

which is independent of θ , as asserted.

The estimator $E_\theta(W|T)$ depends on the actual sample observed in the following sense: after observing the sample $\mathbf{X} = \mathbf{x}^*$, compute $T(\mathbf{x}^*) = t$, and use (11.76) to obtain $E_\theta(W|T)$.

Clearly, if $T(\mathbf{x}^*) = T(\mathbf{x}^{**}) = t$, then the value of the estimator $E_\theta(W|T)$ is the same for samples $\mathbf{x}^*$ and $\mathbf{x}^{**}$. This means that this value depends on the sample only through the value of the sufficient statistic T .

It remains now to prove the inequality (11.75). Based on Theorem 7.7.2,

$$\text{Var}(X) = E_Y\{\text{Var}(X|Y)\} + \text{Var}_Y\{E(X|Y)\}. \quad (11.76)$$

Letting $E_\theta(W) = k(\theta)$, the risk of estimator W is

$$\begin{aligned} R(W, \theta) &= E\{[W - g(\theta)]^2\} = E\{([W - k(\theta)] + [k(\theta) - g(\theta)])^2\} \\ &= E\{[W - k(\theta)]^2\} + [k(\theta) - g(\theta)]^2 \\ &= \text{Var}(W) + [k(\theta) - g(\theta)]^2, \end{aligned} \quad (11.77)$$

since the cross product is easily seen to vanish. The second term is simply the square of the bias [recall that W is used to estimate $g(\theta)$, rather than θ].

We will now use the formula (11.76), taking $X = W$ and $Y = T$. Thus, for every θ we have

$$\begin{aligned} \text{Var}(W) &= E_T\{\text{Var}(W|T)\} + \text{Var}_T\{E(W|T)\} \\ &\geq \text{Var}_T\{E(W|T)\} = \text{Var}(W^*). \end{aligned} \quad (11.78)$$

Now, by the fact that

$$k(\theta) = E(W) = E_T\{E(W|T)\} = E(W^*), \quad (11.79)$$

we can write, using (11.77), (11.78), and (11.79),

$$\begin{aligned} R(W, \theta) &\geq \text{Var}(W^*) + [k(\theta) - g(\theta)]^2 = \text{Var}(W^*) + [E(W^*) - g(\theta)]^2 \\ &= E[W^* - g(\theta)]^2 = R(W^*, \theta). \end{aligned}$$

It remains to determine the conditions for equality in (11.75). It is clear that the equality occurs if and only if $k(\theta) \equiv g(\theta)$ and $E_T\{\text{Var}(W|T)\} = 0$. The first condition means that W is an unbiased estimator of $g(\theta)$. The second condition means that $\text{Var}(W|T) \equiv 0$. That is, for every value t of T , the random variable W is constant; hence, W is a function of T . $\square$

What the Rao–Blackwell theorem says is that in order to decrease the risk of an estimator V , one should decrease its bias. Then, if the estimator is not a function of sufficient statistic (or statistics), replace it by the estimator defined as the conditional expectation of V given sufficient statistics.

We will now illustrate the second part of this principle. Thus, in the examples below we will start from some unbiased estimators and improve them by conditioning with respect to sufficient statistics.

■ EXAMPLE 11.47

Let $X_1, \dots, X_n$ be a random sample from $\text{POI}(\lambda)$ distribution. Suppose that we want to estimate $P\{X = 0\} = e^{-\lambda}$.

We know (see Example 11.42) that the sufficient statistic for λ is $\bar{X}$, or equivalently, $T = \sum_{i=1}^n X_i$, which has $\text{POI}(n\lambda)$ distribution. Let us estimate $P\{X_i = 0\} = e^{-\lambda} = g(\lambda)$ by $W = U/n$ —a relative frequency of X_i 's in the sample that are equal to 0. To compare the risks of W and $W^* = E(W|T)$, it is enough to compare their variances, since it will be shown that W and W^* are both unbiased estimators for $g(\lambda)$.

Clearly, U is binomial with parameters n and $g(\lambda)$, so $E(W) = (1/n)E(U) = g(\lambda)$. Thus, W is unbiased for $g(\lambda)$, and its risk equals

$$R(W, \lambda) = \text{Var}(U/n) = \frac{e^{-\lambda}(1 - e^{-\lambda})}{n}. \quad (11.80)$$

Now let W^* be defined as

$$W^* = E(W|T) = \frac{1}{n}E(U|T). \quad (11.81)$$

Clearly, the number of X_i 's which are equal to 0 cannot exceed n . If $T = 1$, then $U = n - 1$, so it remains to consider cases where $T = t > 1$.

Let us start by determining the conditional distribution of one of the observations, say X_1 , given $T = t$. We have, for $j \leq t$,

$$\begin{aligned} P\left\{X_1 = j \mid \sum_{i=1}^n X_i = t\right\} &= \frac{P\{X_1 = j, \sum_{i=2}^n X_i = t - j\}}{P\{\sum_{i=1}^n X_i = t\}} \\ &= \frac{\frac{\lambda^j e^{-\lambda}}{j!} \times \frac{[(n-1)\lambda]^{t-j}}{(t-j)!} e^{-(n-1)\lambda}}{\frac{(n\lambda)^t}{t!} e^{-n\lambda}} \\ &= \binom{t}{j} \left(\frac{1}{n}\right)^j \left(1 - \frac{1}{n}\right)^{t-j}, \end{aligned}$$

which shows that given $\sum X_i = t$, the variable X_1 is binomial with parameters t and $1/n$.

Now let

$$Y_i = \begin{cases} 1 & \text{if } X_i = 0 \\ 0 & \text{if } X_i > 0, \end{cases}$$

so that for each Y_j we have

$$E(Y_j|T = t) = P(X_j = 0|T = t) = P(X_1 = 0|T = t) = \left(1 - \frac{1}{n}\right)^t.$$

Since $U = Y_1 + \cdots + Y_n$, we obtain

$$\begin{aligned} W^* &= \frac{1}{n}E(U|T) = \frac{1}{n}E(Y_1 + \cdots + Y_n|T) \\ &= \frac{1}{n} \sum_{j=1}^n E(Y_j|T) = \left(1 - \frac{1}{n}\right)^T, \end{aligned} \quad (11.82)$$

a result that is somewhat unexpected.

Finding the risk of W^* (in order to compare it with the risk of W) requires some work. We know that W^* must be unbiased because $W = U/n$ was unbiased, and therefore, we need to compare only variances of W^* and W . Recalling that $T = \sum_{i=1}^n X_i$ has $\text{POI}(n\lambda)$ distribution, we have

$$\begin{aligned} Ez^T &= \sum_{t=0}^{\infty} z^t \times P\{T = t\} = \sum_{t=0}^{\infty} z^t \frac{(n\lambda)^t}{t!} e^{-n\lambda} \\ &= e^{-n\lambda} e^{zn\lambda} \sum_{t=0}^{\infty} \frac{(zn\lambda)^t}{t!} e^{-zn\lambda} = e^{n\lambda(z-1)}. \end{aligned}$$

On the other hand, $E[z^T]^2 = E[(z^2)^T] = e^{n\lambda(z^2-1)}$. Consequently

$$\text{Var}(z^T) = E[z^T]^2 - [E(z^T)]^2 = e^{n\lambda(z^2-1)} - e^{2n\lambda(z-1)}.$$

The estimator W^* is obtained by putting $z = 1 - 1/n$, which gives

$$\begin{aligned} \text{Var}(W^*) &= e^{n\lambda[(n-1)/n]^2 - 1} - e^{2n\lambda[(n-1)/n - 1]} \\ &= e^{-\lambda(2-1/n)} - e^{-2\lambda} = e^{-2\lambda}(e^{\lambda/n} - 1). \end{aligned}$$

To compare the risks (variances) of the original estimator $W = U/n$ and the improved estimator W^* , observe that the risk of W , given by (11.80), can be written as

$$\frac{e^{-\lambda}(1 - e^{-\lambda})}{n} = \frac{\lambda e^{-2\lambda}}{n} + \frac{e^{-2\lambda}}{n}(e^{\lambda} - \lambda - 1).$$

On the other hand, for the risk of W^* we have

$$\begin{aligned} e^{-2\lambda}(e^{\lambda/n} - 1) &= e^{-2\lambda} \left(\frac{\lambda}{n} + \frac{\lambda^2}{2n^2} + \frac{\lambda^3}{6n^3} + \cdots \right) \\ &= \frac{\lambda e^{-2\lambda}}{n} + \frac{e^{-2\lambda}}{n^2} \left(\frac{\lambda e^2}{2} + \frac{\lambda^3}{6n} + \cdots \right), \end{aligned}$$

which is smaller if n is sufficiently large.

■ **EXAMPLE 11.48**

Consider now estimators T_1 – T_5 of θ in the $U[0, \theta]$ distribution introduced in Example 11.5. Three of them, T_3 , T_4 , and T_5 , are not functions of a statistic $T_1 = X_{n:n}$ sufficient for θ (see Example 11.44). Theorem 11.6.3 will be applied to improve estimators T_3 , T_4 , and T_5 .

SOLUTION. The improved estimator T_3 is

$$T_3^* = E_\theta(T_3|T_1) = E_\theta\{(T_1 + X_{1:n})|T_1\} = T_1 + E_\theta(X_{1:n}|T_1).$$

Given T_1 , other $n - 1$ observations are iid, each with a $U[0, T_1]$ distribution. Conditional expectation $E(X_{1:n}|T_1)$ is the same as the expectation of a minimum of $X'_1, \dots, X'_{n-1}$, each X'_i being uniform on $[0, T_1]$. This conditional expectation equals $(1/n)T_1$. Consequently,

$$T_3^* = T_1 + \frac{1}{n}T_1 = \frac{n+1}{n}T_1,$$

which is equal to the estimator T_2 considered in Example 11.5.

Next

$$T_4^* = E_\theta\{(n+1)X_{1:n}|T_1\} = (n+1)\frac{1}{n}T_1,$$

where we used the results obtained in determining T_3^* . Finally, for T_5 we have

$$\begin{aligned} T_5^* &= E_\theta\{T_5|T_1\} = E_\theta\left\{\frac{2}{n}(X_1 + \dots + X_n)|T_1\right\} \\ &= \frac{2}{n}E(X_1 + \dots + X_n|T_1). \end{aligned}$$

In this case, given T_1 , the sum $X_1 + \dots + X_n$ must have one term equal to T_1 , while the others have a uniform distribution on $[0, T_1]$. So again, letting X'_i be uniform on $(0, T_1)$, we have

$$\begin{aligned} E(X_1 + \dots + X_n|T_1) &= T_1 + E(X'_1 + \dots + X'_{n-1}) \\ &= T_1 + (n-1)\frac{T_1}{2} = \frac{n+1}{2}T_1, \end{aligned}$$

and therefore

$$T_5^* = \frac{2}{n} \times \frac{n+1}{2}T_1 = \frac{n+1}{n}T_1.$$

Observe that improving three different estimators (T_3 , T_4 , and T_5) by conditioning with respect to sufficient statistic T_1 led in each case to the same unbiased estimator $T_2 = [(n+1)/n]T_1$. The natural question arises: Was it a coincidence, or a rule? If such a result occurs as a rule, we could always select an unbiased estimator for which the conditioning would be especially simple. Thus, we would find the unique unbiased estimator that could not be improved any further, hence with minimal risk. We will now introduce one more concept that connects all the pieces together.

The situation at present may be summarized as follows: On the one hand, we have the Cramér–Rao inequality, which tells us how much reduction of the variance (hence of the risk for unbiased estimators) is possible. This bound may be effectively calculated in textbook cases, but in real-life situations, it can be hard or impossible to determine.

On the other hand, we have the Rao–Blackwell theorem, which tells us that unbiased estimators that are not based on sufficient statistics can be improved; to be more specific, their variance can be reduced by conditioning on sufficient statistics.

Again, the technical difficulties of such conditioning can be formidable. But here the situation can often be simplified. Rather than try to improve an estimator by finding its conditional expectation on a sufficient statistic, we can simply abandon the effort and refrain from using estimators that are not functions of sufficient statistics. Instead, we can try to find an unbiased estimator that is built only on sufficient statistics. This may still be technically difficult but often easier than determining conditional expectations.

Once we find an unbiased estimator that is a function of a sufficient statistic (or a minimal sufficient set of statistics), the Rao–Blackwell theorem tells us that it cannot be improved any further (at least by conditioning). We therefore are tempted to conclude that we found the MVUE. Such a conclusion is justified if the estimator in question has variance equal to the right-hand side of the Cramér–Rao inequality. Otherwise, the reasoning on which it is based still has a flaw and has to be amended. Indeed, this reasoning works if there is only one unbiased estimator based on a minimal sufficient statistic (or minimal sufficient set of statistics). But what if there are several such estimators? We know that none of the estimators can be improved by further conditioning, but this does not guarantee that their risks are all minimal.

This observation makes it necessary to single out the cases where such an “anomaly” cannot occur, that is, situations where an unbiased estimator based on a sufficient statistic is unique.

Accordingly, we introduce the following definition:

Definition 11.6.5 A family of distributions $\{f(t; \theta), \theta \in \Theta\}$ of variable T is called *complete* if for all $\theta \in \Theta$ the condition $E_\theta[u(T)] = 0$ implies that $u(T) = 0$ with probability 1. A sufficient statistic with a distribution in a complete family is said to be a *complete sufficient* statistic. $\square$

We have the following theorem:

Theorem 11.6.4 If the family of distributions $\{f(t; \theta), \theta \in \Theta\}$ is complete, then any unbiased estimator $u(T)$ of parameter $g(\theta)$ is unique.

Proof: If $u_1(T)$ and $u_2(T)$ are unbiased estimators of $g(\theta)$, then

$$E_\theta[u_1(T)] = E_\theta[u_2(T)] = g(\theta),$$

and $E_\theta\{u_1(T) - u_2(T)\} = 0$ for all $\theta \in \Theta$. Hence $P_\theta\{u_1(T) - u_2(T) = 0\} = 1$ for all θ , which means that the functions u_1 and u_2 coincide. $\square$

Theorem 11.6.5 Lehmann and Scheffé Let $X_1, \dots, X_n$ be a random sample from a distribution $\{f(x; \theta), \theta \in \Theta\}$, and let $\{T_1, \dots, T_k\}$ be jointly complete sufficient statistics for θ . If the statistic $U^* = U^*(T_1, \dots, T_k)$ is an unbiased estimator of $g(\theta)$, then U^* is the minimum variance unbiased estimator (MVUE) of $g(\theta)$.

Proof: By completeness, any statistic $U = U(T_1, \dots, T_k)$ that is unbiased for $g(\theta)$ must satisfy the condition $U = U^*$ with probability 1 (for every θ). On the other hand, for any statistic $W = W(X_1, \dots, X_n)$ that is unbiased for $g(\theta)$ and is not a function of $T_1, \dots, T_k$, $U = E\{W|T_1, \dots, T_k\}$ will also be unbiased and will be a function of $T_1, \dots, T_k$ such that $\text{Var}_\theta(U) \leq \text{Var}_\theta(W)$, which shows that variance of U^* is minimal for each θ . $\square$

The concept of completeness pertains to a family of distributions. It is usually stated in a phrase appealing to intuition and easy to remember: A family is complete if “there are

no nontrivial unbiased estimators of zero" (a trivial unbiased estimator of zero is a random variable identically equal to zero). We will now give some examples of complete families.

■ EXAMPLE 11.49

Consider random sampling from a $\text{POI}(\lambda)$ distribution. A sufficient statistic for λ , $T = \sum_{i=1}^n X_i$, has $\text{POI}(n\lambda)$ distribution. Let $u(T)$ be a statistic, and suppose that $E_\theta[u(T)] = 0$ for all θ . This means that

$$E_\theta[u(T)] = \sum_{n=0}^{\infty} u(n) \frac{\theta^n}{n!} e^{-\theta} = e^{-\theta} \sum_{n=0}^{\infty} u(n) \frac{\theta^n}{n!} = 0.$$

Since $e^{-\theta} \neq 0$, we obtain $g(\theta) = \sum u(n)\theta^n/n! = 0$ for all θ . However, since the Taylor expansion of $g(\theta)$ is $\sum [g^{(n)}(0)/n!]\theta^n$, we have $u(n) = g^{(n)}(0)$. But the latter quantity is zero for all n .

Observe here that we used the fact that the family contains all Poisson distributions as indexed by θ (or at least sufficiently many of them). If we consider the family of some selected Poisson distributions, say for $\theta_1, \theta_2, \dots, \theta_r$, we can choose some negative and some positive values of $u(t)$ so as to obtain appropriate cancelations. The simplest example here is to take a family consisting of just one Poisson distribution, with mean $\theta = 5$ (say). Then, the function $u(T) = T - 5$ is not identically zero, yet its expectation is zero for all members of the family (since there is only one distribution in the family, and the expectation of this distribution is 5).

■ EXAMPLE 11.50

In a $U[0, \theta]$ distribution, the sufficient statistic for θ is $T_1 = X_{n:n}$, and its density $f(t, \theta)$ is

$$f(t; \theta) = \begin{cases} nt^{n-1}/\theta^n & 0 \leq t \leq \theta \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that we have a function u that satisfies the condition

$$E_\theta[u(T)] = \int_0^\theta \frac{u(t)nt^{n-1}}{\theta^n} dt = 0.$$

This means that

$$\int_0^\theta t^{n-1}u(t) dt = 0,$$

and by differentiation with respect to θ , we get $\theta^{n-1}u(\theta) = 0$ for all $\theta > 0$. Consequently, we must have $u(t) \equiv 0$, and therefore, the class of all distributions uniform on $[0, \theta]$, $\theta > 0$, is complete.

This explains why in Example 11.48, by conditioning the unbiased estimators T_3, T_4 , and T_5 on sufficient statistic T_1 , we obtain the (unique, in view of completeness) unbiased estimator $[(n+1)/n]T_1$ of θ .

Proving completeness of a family starting directly from the definition may not be easy. We will therefore present a rather general and easily verifiable sufficient condition for completeness:

Definition 11.6.6 The distribution $f(x, \theta)$ is in an *exponential class*, if

$$f(x; \theta) = \begin{cases} c(\theta)b(x) \exp \left[\sum_{i=1}^m q_i(\theta)k_i(x) \right] & \text{if } x \in A \\ 0 & \text{otherwise,} \end{cases} \quad (11.83)$$

where $\theta = (\theta_1, \dots, \theta_m)$ and $c(\theta), b(x)$ are strictly positive functions. It is assumed here that the parameter space is a generalized rectangle (possibly infinite in some dimensions) of the form $\Theta = \{\theta : a_i \leq \theta_i \leq b_i, i = 1, \dots, m\}$.

Moreover,

1. The set A where $f(x; \theta)$ is positive does not depend on θ .
2. The functions $q_1(\theta), \dots, q_m(\theta)$ are continuous and functionally independent (i.e., none is a function of others).
3. The functions $k_1(x), \dots, k_m(x)$ are linearly independent (i.e., no linear combination of them is identically zero unless all coefficients are zero), and when $f(x; \theta)$ is a density of a continuous random variable, they are differentiable.

□

We have the following theorem:

Theorem 11.6.6 Let $X_1, \dots, X_n$ be a random sample from a distribution $\{f(x, \theta), \theta \in \Theta\}$ that belongs to the exponential class (11.83). Then the set of statistics $\{T_1, \dots, T_m\}$ such that

$$T_j(X_1, \dots, X_n) = \sum_{i=1}^n k_j(X_i), \quad \text{for } j = 1, \dots, m$$

is a minimal set of complete sufficient statistics for $\theta = (\theta_1, \dots, \theta_m)$.

The joint sufficiency of statistics $T_1, \dots, T_m$ follows immediately from (11.83). For the proof of completeness, see Lehmann and Romano (2005).

Distributions from most common families (e.g., binomial, Poisson, normal, or gamma) belong to the exponential class. We will show it for the binomial and normal distributions, leaving the proofs for the remaining families as exercises.

■ EXAMPLE 11.51

For the Bernoulli distribution, we have for $x = 0, 1$

$$\begin{aligned} f(x; p) &= p^x(1-p)^{1-x} = (1-p) \left(\frac{p}{1-p} \right)^x \\ &= (1-p) \exp \left\{ x \log \left(\frac{p}{1-p} \right) \right\}. \end{aligned}$$

Thus, $A = \{0, 1\}$, $m = 1$, $q_1(p) = \log[p/(1-p)]$, and $k_1(x) = x$. The sufficient statistic is $T = \sum k_1(X_i) = \sum X_i$.

■ EXAMPLE 11.52

For a normal distribution we have, after letting $\theta = (\mu, \sigma)$,

$$\begin{aligned} f(x, \theta) &= \frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\} = \frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ -\frac{(x^2 - 2\mu x + \mu^2)}{2\sigma^2} \right\} \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ \frac{-\mu^2}{2\sigma^2} \right\} \exp \left\{ -\frac{1}{2\sigma^2}x^2 + \frac{\mu}{\sigma^2}x \right\}, \end{aligned}$$

so we have $c(\theta) = (1/\sigma\sqrt{2\pi})e^{-\mu^2/2\sigma^2}$, $b(x) \equiv 1$, $q_1(\theta) = -1/2\sigma^2$, $q_2(\theta) = \mu/\sigma^2$, $k_1(x) = x^2$, $k_2(x) = x$. A minimal sufficient set of statistics is therefore $T_1 = \sum_{i=1}^n X_i^2$ and $T_2 = \sum_{i=1}^n X_i$.

There exist families of distributions in which the dimension of parameter θ is less than the number of jointly sufficient statistics. Such families are called *curved exponential*. Most properties of their properties are the same as the properties of so-called full exponential families.

■ EXAMPLE 11.53

Let us consider a $N(\theta, \theta^2)$ distribution. Parameter θ is now one-dimensional and the density is

$$f(x; \theta) = \frac{1}{\sqrt{2\pi}\theta} e^{-(x-\theta)^2/(2\theta^2)} = \frac{1}{\sqrt{2\pi}\theta} e^{-x^2/(2\theta^2) + x/\theta - 1/2}.$$

Statistics $T_1 = \sum X_i^2$ and $T_2 = \sum X_i$ are jointly minimal sufficient for $(\mu, \sigma^2) = (\theta, \theta^2)$. Parameters space Θ is a parabola.

■ EXAMPLE 11.54

In this example, the family of distributions is $\text{GAM}(\alpha, 1/\alpha)$ with density

$$f(x; \alpha) = \frac{\alpha^{-\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-x/\alpha} = \frac{\alpha^{-\alpha}}{\Gamma(\alpha)} e^{(\alpha-1) \log x - x/\alpha}.$$

Parameter θ is also one-dimensional, and statistics $T_1 = \sum \log X_i^2$ and $T_2 = \sum X_i$ are jointly minimal sufficient for $(\alpha, \lambda) = (\alpha, 1/\alpha)$. Parameters space Θ is a hyperbola.

We end this section with one more concept whose notion is complementary to the notion of a sufficient statistic.

Definition 11.6.7 Let $X_1, \dots, X_n$ be a random sample from a distribution $f(x, \theta)$. Statistic $T = T(X_1, \dots, X_n)$ is *ancillary* for parameter θ , if its distribution does not depend on θ . $\square$

■ EXAMPLE 11.55

In a random sample of size n obtained from the $N(\theta, 1)$ distribution, statistic $T = \sum_{i=1}^n (X_i - \bar{X})^2$ has a chi-square distribution with $n - 1$ degrees of freedom, so it is ancillary for θ .

The following theorem, due to Basu, establishes the independence of complete sufficient and ancillary statistics:

Theorem 11.6.7 Basu Let $X_1, \dots, X_n$ be a random sample from the distribution $\{f(x, \theta), \theta \in \Theta\}$. If $\{T_1, \dots, T_m\}$ are jointly complete sufficient statistics for θ , then $\{T_1, \dots, T_m\}$ and any ancillary statistic $V = V(X_1, \dots, X_n)$ are independent.

PROBLEMS

- 11.6.1** Find sufficient statistics (or show that they do not exist) for parameter θ in the following distributions: (i) $f(x, \theta) = (x/\theta^2)e^{-x^2/2\theta^2}$ for $x > 0$ (Rayleigh). (ii) $f(x, \theta) = (1/2\theta)e^{-|x|^\theta}$ (double exponential). (iii) $\text{BETA}(\theta, 2\theta)$. (iv) $\text{U}[\theta, 2\theta]$. (v) $\text{GAM}(\theta, \theta)$. (vi) $f(x, \theta) = \{\pi[1 + (x - \theta)^2]\}^{-1}$ (Cauchy).

- 11.6.2** Generalizing Example 11.39, let $X_1, \dots, X_n$ be n independent Bernoulli trials. Show that $T = \sum_{i=1}^n X_i$ is sufficient for probability of success p by finding the joint distribution of $(X_1, \dots, X_n)$ given $T = t$. Find the marginal distribution $P\{X_1 = 1|T = t\}$.
- 11.6.3** Find a sufficient statistic for θ if observations are uniformly distributed on the set of integers $0, 1, \dots, \theta$.
- 11.6.4** Show that the $N(0, \theta)$ family is not complete.
- 11.6.5** Let $X_1, \dots, X_n$ be a random sample from the distribution with a density $f(x; \lambda, \theta) = \lambda e^{-\lambda(x-\theta)}$ for $x \geq \theta$ and 0 otherwise. Determine a pair of jointly sufficient statistics for parameters λ and θ .
- 11.6.6** Check if the following families of distributions are in the exponential class: **(i)** $\text{POI}(\lambda)$. **(ii)** $\text{EXP}(\lambda)$. **(iii)** $\text{NBIN}(r, \theta)$, r known. **(iv)** $\text{BETA}(\theta_1, \theta_2)$. **(v)** $\text{WEI}(k, \theta)$, k known. **(vi)** $\text{WEI}(\theta, /)$, $/$ known.
- 11.6.7** Show that the family of $\text{GAM}(\alpha, \lambda)$ distributions is in the exponential class, and find the minimal jointly sufficient statistics.
- 11.6.8** Suppose that a random sample is taken from a distribution with density $f(x; \theta) = 2x/\theta^2$ for $0 \leq x \leq \theta$ and $f(x; \theta) = 0$ otherwise. Find the MLE of the median of this distribution, and show that this estimator is a minimal sufficient statistic.
- 11.6.9** Let $X_1, \dots, X_n$ be a random sample from the $\text{EXP}(\lambda)$ distribution. Suppose that only first k order statistics $X_{1:n}, \dots, X_{k:n}$ are observed. Find a minimal sufficient statistic for λ .
- 11.6.10** Let X_1, X_2 be a random sample of size 2 from the $\text{POI}(\lambda)$ distribution. Show that statistic $T = X_1 + 2X_2$ is not sufficient for λ .
- 11.6.11** Let $X_1, \dots, X_n$ be a random sample from the $U(\theta, \theta + 1)$ distribution. Show that statistic $T = X_{n:n} - X_{1:n}$ is ancillary for θ .
- 11.6.12** Let X_1, X_2 be a random sample of size $n = 2$ from the $\text{EXP}(\lambda)$ distribution. Show that statistic $T = X_1/X_2$ is ancillary for λ .
- 11.6.13** Let $X_1, \dots, X_n$ be a random sample from the $U(0, \theta)$ distribution. Show that statistic $T = X_{n:n}/X_{1:n}$ is ancillary for θ .

11.7 INTERVAL ESTIMATION

Thus far, we have dealt with point estimation. An estimator was used to produce a single number, hopefully close to an unknown parameter. The natural question arises: What can be said about the distance between the value produced by the estimator and the true value of the parameter?

Confidence Intervals

If the value of θ is fixed then all randomness is in the process of sampling and one cannot speak of “the probability that the value of parameter θ belongs to some set.” The relevant statements on interval estimation are expressed through the concept of *confidence intervals* (CI).

Let $\mathbf{X} = (X_1, \dots, X_n)$ be a random sample from the distribution $f(x, \theta)$. Let α be a fixed number with $0 < \alpha < 1$ (typically α is a small number such as $\alpha = 0.05$ or $\alpha = 0.01$).

Definition 11.7.1 A pair of statistics $L = L(\mathbf{X})$ and $U = U(\mathbf{X})$ is an $(1 - \alpha)$ -level confidence interval (CI) for θ if for all $\theta \in \Theta$,

$$P_{\theta}\{L(\mathbf{X}) \leq \theta \leq U(\mathbf{X})\} = 1 - \alpha. \quad (11.84)$$

Similarly, a statistic $L(\mathbf{X})$ is a $(1 - \alpha)$ -level lower confidence bound if

$$P\{L(\mathbf{X}) \leq \theta\} = 1 - \alpha. \quad (11.85)$$

□

The upper confidence bound $U(\mathbf{X})$ is defined in a similar way.

The interval $[L(\mathbf{X}), U(\mathbf{X})]$ varies from sample to sample and is therefore random. By definition, the probability that it covers the unknown (fixed) value of θ is $1 - \alpha$. This probability, however, refers to the process of taking the sample only. After a specific sample $\mathbf{x}$ has been observed, the observed interval $[L(\mathbf{x}), U(\mathbf{x})]$ either covers or does not cover θ . All randomness is gone, and it makes no sense to speak of “the probability that θ belongs to the interval $[L(\mathbf{x}), U(\mathbf{x})]$ ” because this probability is now either 0 or 1. Since we do not know which is the case, the term *confidence* is used. We say that the $(1 - \alpha)$ -level confidence interval for θ is $[L(\mathbf{x}), U(\mathbf{x})]$. In frequentist terms, the last statement means “the confidence interval $[L(\mathbf{x}), U(\mathbf{x})]$ has been obtained by a method that produces randomly generated intervals, of which, on average, $100(1 - \alpha)\%$ cover the unknown value θ .”

It is perhaps regrettable that the terminology here, as it developed historically, is a little ambiguous. The term *confidence interval* refers to interval $L[(\mathbf{X}), U(\mathbf{X})]$ with random endpoints as well as to the interval $L[(\mathbf{x}), U(\mathbf{x})]$ with endpoints obtained from the sample observed. Sometimes, the first interval is called the *probability* interval, and the second is called the *sample* confidence interval. We will follow a more common tradition, referring to confidence intervals and sample confidence intervals.

In constructing confidence intervals, we use quantiles of some known distributions, such as standard normal, Student's t , and chi-square. For typographical reasons, as well as to follow the tradition, we use different notation for distributions symmetric about 0 ($N(0, 1)$ and Student's t).

Thus, if Z is a standard normal random variable, then z_{α} will denote the $(1 - \alpha)$ -quantile of Z (or, upper α -quantile), that is,

$$P\{Z \leq z_{\alpha}\} = 1 - \alpha. \quad (11.86)$$

Then, $z_{1-\alpha} = -z_{\alpha}$, and $P\{|Z| \leq z_{\alpha/2}\} = 1 - \alpha$. Similarly, the $(1 - \alpha)$ -quantile of Student's t distribution with ν degrees of freedom will be denoted by $t_{\alpha, \nu}$.

For asymmetric distributions, such as chi-square, the subscript α will be used for upper quantiles of order α so that $\chi_{\alpha, \nu}^2$ denotes the quantile of order $1 - \alpha$ of a chi-square distribution with ν degrees of freedom.

■ EXAMPLE 11.56

Consider the random sample $X_1, \dots, X_n$ from the distribution $N(\theta, \sigma^2)$, where θ is the unknown parameter while σ^2 is known. Then, the interval

$$\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

is a $(1 - \alpha)$ -level confidence interval for θ . Indeed, the inequality

$$\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \theta < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

is equivalent to

$$\theta - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} < \theta + z_{\alpha/2} \frac{\sigma}{\sqrt{n}},$$

so both inequalities have the same probabilities. Since $\text{Var}(\bar{X}) = \sigma^2/n$, we have

$$\begin{aligned} P \left\{ \bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \theta < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right\} \\ = P \left\{ -z_{\alpha/2} < \frac{\bar{X} - \theta}{\sigma/\sqrt{n}} < z_{\alpha/2} \right\} = 1 - \alpha, \end{aligned}$$

which was to be shown.

■ EXAMPLE 11.57

Let $X_1, \dots, X_n$ be a random sample from a continuous distribution with density $f(x, \theta)$ depending on a one-dimensional parameter θ . Furthermore, let $F(x, \theta)$ be the cdf of X , and let $G = 1 - F$. Then, each of the random variables $-2 \log F(X, \theta)$ and $-2 \log G(X, \theta)$ has an exponential distribution with parameter $1/2$. Indeed

$$P\{-2 \log F(X, \theta) > t\} = P\{F(X, \theta) < e^{-t/2}\} = e^{-t/2},$$

since $F(X, \theta)$ is uniform on $[0, 1]$. The argument for $G(X, \theta)$ is analogous. Consequently, the sums

$$U = -2 \sum_{i=1}^n \log F(X_i, \theta) \quad \text{and} \quad W = -2 \sum_{i=1}^n \log G(X_i, \theta) \quad (11.87)$$

have $\text{GAM}(n, 1/2)$ distribution, which is the same as chi-square distribution with $2n$ degrees of freedom. Therefore,

$$P \left\{ \chi_{1-\alpha/2, 2n}^2 < -2 \sum_{i=1}^n \log F(X_i, \theta) < \chi_{\alpha/2, 2n}^2 \right\} = 1 - \alpha \quad (11.88)$$

and

$$P \left\{ \chi_{1-\alpha/2, 2n}^2 < -2 \sum_{i=1}^n \log G(X_i, \theta) < \chi_{\alpha/2, 2n}^2 \right\} = 1 - \alpha. \quad (11.89)$$

If the inequality that defines the event in either (11.88) or ((11.89)) can be solved to the form $L(X_1, \dots, X_n) < \theta < U(X_1, \dots, X_n)$, we obtain, by definition, a confidence interval for θ at level $1 - \alpha$.

■ EXAMPLE 11.58

Continuing Example 11.57, if $X_1, \dots, X_n$ have an exponential distribution with density $\theta e^{-\theta x}$, then $F(x, \theta) = 1 - e^{-\theta x}$ and $G(x, \theta) = e^{-\theta x}$. Formula (11.89) gives

$$P \left\{ \chi_{1-\alpha/2, 2n}^2 < 2\theta \sum_{i=1}^n X_i < \chi_{\alpha/2, 2n}^2 \right\} = 1 - \alpha,$$

and we obtain the $(1 - \alpha)$ -level confidence interval

$$\frac{\chi_{1-\alpha/2, 2n}^2}{2n\bar{X}} < \theta < \frac{\chi_{\alpha/2, 2n}^2}{2n\bar{X}}. \quad (11.90)$$

Note that if we used F instead of G , we would obtain the inequality

$$\chi_{1-\alpha/2, 2n}^2 < -2 \sum_{i=1}^n \log(1 - e^{-\theta X_i}) < \chi_{\alpha/2, 2n}^2,$$

which cannot easily be solved for θ . Thus, even when we have an explicit formula for F (and hence for G), use of one of these functions often leads to simpler results than use of the other.

In general, how are confidence intervals obtained? For large n , the answer lies in the limit Theorem 11.5.1 about the asymptotic distributions of MLE's obtained from setting the derivative of the likelihood equal to 0 and solving the resulting equation. Indeed, since $\sqrt{nI(\theta)}(\hat{\theta}_n - \theta)$ with $I(\theta)$ defined by (11.27) is asymptotically standard normal (as in Example 11.56), we have

$$-z_{\alpha/2} < \sqrt{nI(\theta)}(\hat{\theta}_n - \theta) < z_{\alpha/2}$$

with probability close to $1 - \alpha$. This inequality is equivalent to the inequality

$$\hat{\theta}_n - \frac{z_{\alpha/2}}{\sqrt{nI(\theta)}} < \theta < \hat{\theta}_n + \frac{z_{\alpha/2}}{\sqrt{nI(\theta)}}. \quad (11.91)$$

The latter is not a confidence interval, since the bounds depend on the unknown value of θ . However, for large n , we can replace θ by $\hat{\theta}_n$ in $I(\theta)$. Consequently, we can state the following theorem:

Theorem 11.7.1 *Let $\hat{\theta}_n$ be the MLE estimator of θ in a problem for which MLE's satisfy assumption of Theorem 11.5.1. Then for large n ,*

$$\left(\hat{\theta}_n - \frac{z_{\alpha/2}}{\sqrt{nI(\hat{\theta}_n)}}, \hat{\theta}_n + \frac{z_{\alpha/2}}{\sqrt{nI(\hat{\theta}_n)}} \right)$$

is an approximate $(1 - \alpha)$ -level confidence interval for parameter θ .

■ EXAMPLE 11.59

As an illustration, we consider the problem of estimating probability p in Bernoulli distribution. The observations $X_1, X_2, \dots, X_n$ are iid random variables indicating whether consecutive trials lead to success or failure (i.e., $X_i = 1$ or 0 with probabilities p and $1 - p$). The sum $S_n = X_1 + \dots + X_n$ is the total number of successes in n trials. The MLE of p is $\hat{p} = S_n/n$, and (11.35) gives $I(p) = 1/p(1 - p)$ so that

$$\text{Var}(\hat{p}) = \frac{1}{n^2} \text{Var}(S_n) = \frac{p(1 - p)}{n} = \frac{1}{nI(p)}.$$

By the central limit theorem for the binomial distribution, the random variable

$$\frac{S_n - np}{\sqrt{np(1 - p)}} = \sqrt{nI(p)}(\hat{p} - p)$$

is asymptotically standard normal. Consequently, we have the approximate relation

$$P \left\{ -z_{\alpha/2} < \sqrt{nI(p)}(\hat{p} - p) < z_{\alpha/2} \right\} = 1 - \alpha. \quad (11.92)$$

The inequality on the left-hand side of (11.92) is equivalent to

$$|\hat{p} - p| < \frac{z_{\alpha/2}}{\sqrt{n}} \sqrt{p(1 - p)}$$

or

$$\left(1 + \frac{z_{\alpha/2}^2}{n}\right) p^2 - \left(2\hat{p} + \frac{z_{\alpha/2}^2}{n}\right) p + \hat{p}^2 < 0. \quad (11.93)$$

The solution of the quadratic inequality (11.93),

$$\frac{\hat{p} + z_{\alpha/2}^2/2n - z_{\alpha/2} \sqrt{\frac{4\hat{p}(1-\hat{p}) + z_{\alpha/2}^2/n}{4n}}}{1 + z_{\alpha/2}^2/n} < p < \frac{\hat{p} + z_{\alpha/2}^2/2n + z_{\alpha/2} \sqrt{\frac{4\hat{p}(1-\hat{p}) + z_{\alpha/2}^2/n}{4n}}}{1 + z_{\alpha/2}^2/n}, \quad (11.94)$$

gives an approximate $(1 - \alpha)$ -level confidence interval for probability p .

If we disregard the terms of the order higher than $1/\sqrt{n}$, we obtain a more commonly used further approximation, namely

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < p < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \quad (11.95)$$

which coincides with the confidence interval given in Theorem 11.7.1

The method of obtaining confidence intervals for any n is based on the pivotal variables defined as follows.

Definition 11.7.2 A random variable W is called *pivotal* for θ if it depends only on the sample $\mathbf{X}$ and on unknown parameter θ , while its distribution does not depend on any unknown parameters (including θ). $\square$

This means that in principle one should be able to calculate the numerical values of probabilities for W . It is not enough to know that W has, for example, a normal distribution with one or more parameters unspecified.

If $W = W(\mathbf{X}, \theta)$ is a pivotal random variable, then the $(1 - \alpha)$ -level confidence interval can be obtained as follows: First, for given α , one determines the values q_{α}^* and q_{α}^{**} such that

$$P\{q_{\alpha}^* \leq W(\mathbf{X}, \theta) \leq q_{\alpha}^{**}\} = 1 - \alpha, \quad (11.96)$$

which is possible since W is pivotal.

It now remains to convert the inequality $q_{\alpha}^* \leq W(\mathbf{X}, \theta) \leq q_{\alpha}^{**}$ into the form

$$L(\mathbf{X}, q_{\alpha}^*, q_{\alpha}^{**}) \leq \theta \leq U(\mathbf{X}, q_{\alpha}^*, q_{\alpha}^{**}) \quad (11.97)$$

or, more generally, into the form

$$\theta \in S(\mathbf{X}^{(n)}, q_{\alpha}^*, q_{\alpha}^{**}), \quad (11.98)$$

where S is some set (not necessarily an interval). But as is often the case, if the pivotal quantity W is monotone in θ for every value of $\mathbf{X}$, a solution of the form (11.97) is attainable, and then the statistics L and U provide the lower and upper endpoints of the $(1 - \alpha)$ -level confidence interval for θ .

■ EXAMPLE 11.60

The random variables (11.87) from Example 11.57 are pivotal, which shows that in a continuous case with a one-dimensional parameter there always exist at least two pivotal variables.

The next example shows that pivotal variables can also exist in the case of multidimensional parameters.

■ EXAMPLE 11.61

As in Example 11.56, assume that $X_1, \dots, X_n$ is a random sample from the distribution $N(\theta, \sigma^2)$, but σ^2 is now unknown. We are still interested in estimating the mean θ . In such cases, the statisticians use the term *nuisance* parameter for σ^2 .

In Example 11.56, the random variable $(\bar{X} - \theta)/(\sigma/\sqrt{n})$ was *pivotal*, since σ was known. However, if σ is unknown, we have to proceed differently. The idea is to cancel σ in the denominator of $(\bar{X} - \theta)/(\sigma/\sqrt{n})$.

Recall that when $X_i \sim N(\theta, \sigma^2)$ and $S^2 = (1/n) \sum_{i=1}^n (X_i - \bar{X})^2$, the random variable nS^2/σ^2 has a chi-square distribution with $n - 1$ degrees of freedom (Theorem 9.2.1). Moreover, S^2 is independent of $\bar{X}$. Consequently, the ratio

$$\begin{aligned} T &= \frac{(\bar{X} - \theta)/(\sigma/\sqrt{n})}{\sqrt{nS^2/\sigma^2(n-1)}} = \frac{\bar{X} - \theta}{S} \sqrt{n-1} \\ &= \frac{\bar{X} - \theta}{\sqrt{(1/n) \sum (X_i - \bar{X})^2}} \sqrt{n-1} \end{aligned} \quad (11.99)$$

has the Student's t distribution with $n - 1$ degrees of freedom. Thus, (11.99) is a pivotal random variable (observe that σ is canceled). If now $t_{\alpha/2, n-1}$ is the $(1 - \alpha/2)$ -quantile of Student's t distribution with $n - 1$ degrees of freedom, then we have

$$P \left\{ -t_{\alpha/2, n-1} \leq \frac{\bar{X} - \theta}{S} \sqrt{n-1} \leq t_{\alpha/2, n-1} \right\} = 1 - \alpha,$$

which gives

$$\left(\bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n-1}}, \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n-1}} \right) \quad (11.100)$$

as a $(1 - \alpha)$ -level confidence interval for θ .

■ EXAMPLE 11.62 Confidence Intervals for Variance

Consider again a random sample with normal distribution $N(\mu, \sigma^2)$, this time assuming that the parameter to be estimated is σ^2 . We distinguish two cases: when μ is known, and when μ is unknown. To provide some motivation for considering these cases, consider the problem of assessing the quality of some measuring device. A typical procedure here consists of repeating the measurements of the same object. Assuming that the measurement errors are normal, the two cases above correspond to measurements of an object whose true value μ of the measured attribute is otherwise known, and measurements of an object whose true value μ of the measured attribute is unknown.

The pivotal random variables are respectively,

$$U = \sum_{i=1}^n (X_i - \mu)^2 / \sigma^2 \quad \text{and} \quad V = \sum_{i=1}^n (X_i - \bar{X})^2 / \sigma^2.$$

Since U is a sum of n independent squares of standard normal random variables, it has a chi-square distribution with n degrees of freedom. We know also from Theorem 9.2.1 that V has a chi-square distribution with $n - 1$ degrees of freedom. Thus,

$$P \left\{ \chi_{1-\alpha/2,n}^2 < \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} < \chi_{\alpha/2,n}^2 \right\} = 1 - \alpha$$

and

$$P \left\{ \chi_{1-\alpha/2,n-1}^2 < \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} < \chi_{\alpha/2,n-1}^2 \right\} = 1 - \alpha.$$

We obtain therefore the $(1 - \alpha)$ -level confidence intervals

$$\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{\alpha/2,n}^2} < \sigma^2 < \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{1-\alpha/2,n}^2} \quad (11.101)$$

and

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\chi_{\alpha/2,n-1}^2} < \sigma^2 < \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\chi_{1-\alpha/2,n-1}^2}. \quad (11.102)$$

Two obvious questions arise that concern the intervals (11.101) and (11.102). First, when μ is unknown, one *must* use the interval (11.102). But when μ is known, one can either utilize this knowledge and use interval (11.101) or not utilize it and use interval (11.102). Which is the proper procedure? The answer is that a shorter interval is obtained, on average, if one uses the available information about the expectation μ .

Second, to get the probability $1 - \alpha$ of covering the unknown value of σ^2 , it is not necessary to choose values $\chi_{1-\alpha/2,n}^2$ and $\chi_{\alpha/2,n}^2$, which cut equal probabilities at both tails of the chi-square distribution. One could choose two other points a and b such that the probability of a chi-square random variable assuming a value between a and b is $1 - \alpha$. There are infinitely many such pairs (a, b) , the pair $(\chi_{1-\alpha/2,n}^2, \chi_{\alpha/2,n}^2)$ being just one of them. An obvious criterion would be to choose a pair a, b that minimizes the length of the confidence interval. It is easy to see that this length is proportional to $1/a - 1/b$. We have a similar discussion when μ is unknown.

To solve this optimization problem, let g_m be the density of the chi-square distribution with m degrees of freedom (in our case, we use either $m = n$ or $m = n - 1$). Then, the problem can be formulated as

$$\text{minimize} \quad \left(\frac{1}{a} - \frac{1}{b} \right) \quad \text{subject to} \quad \int_a^b g_m(x) dx \geq 1 - \alpha.$$

This problem can be solved by using Lagrange multipliers: Differentiating the function

$$\frac{1}{a} - \frac{1}{b} + \lambda \left[\int_a^b g_m(x) dx - (1 - \alpha) \right]$$

with respect to a, b , and λ , we obtain three equations:

$$-\frac{1}{a^2} - \lambda g_m(a) = 0, \quad \frac{1}{b^2} + \lambda g_m(b) = 0, \quad (11.103)$$

and

$$\int_a^b g_m(x) dx = 1 - \alpha. \quad (11.104)$$

As a solution choose a, b satisfying constraint (11.104) and such that $a^2 g_m(a) = b^2 g_m(b)$. These can be solved numerically for each m . The solution, for large m , is close to the “symmetric” solution, where $a = \chi_{1-\alpha/2, m}^2$, $b = \chi_{\alpha/2, m}^2$. For small sample sizes, it is better to use an exact solution of the optimization problem above, as it gives the shortest possible confidence interval.

Table B1 gives the optimal left cutoff probability α_1 , and the corresponding upper quantiles $a = \chi_{\alpha_1, m}^2$ and $b = \chi_{\alpha_2, m}^2$, where $\alpha_2 = (1 - \alpha) - \alpha_1$, for $\alpha = 0.1$ and $\alpha = 0.05$ and various numbers of degrees of freedom. The table also gives the relative gain resulting from using the shortest confidence intervals (with length proportional to $1/a - 1/b$), as compared to the confidence interval with equal cutoff probabilities, hence with length proportional to $1/\chi_{1-\alpha/2, m}^2 - 1/\chi_{\alpha/2, m}^2$.

■ EXAMPLE 11.63

Suppose that five observations from a normal distribution with unknown μ and σ^2 are such that $\sum (X_i - \bar{X})^2 = c$. To obtain a 95%CI for σ^2 , the usual procedure based on cutting off 2.5% at both ends of the chi-square distribution with 4 degrees of freedom uses thresholds $\chi_{0.975, 4}^2 = 0.484$ and $\chi_{0.025, 4}^2 = 11.143$. This leads to the confidence interval $c/11.143 < \sigma^2 < c/0.484$, or $(0.090c, 2.066c)$, of length $1.976c$.

The shortest 95% confidence interval involves the upper quantiles $a = \chi_{1-0.0497, 4}^2 = 0.708$ and $b = \chi_{1-0.0497-0.95, 4}^2 = 21.047$. The confidence interval is now $0.0475c = c/21.047 < \sigma^2 < c/0.708 = 1.412c$. The length is $1.3649c$, which is 69% of the length of the interval based on quantiles corresponding to 2.5% probability on each end.

Two natural questions arise: Do pivotal random variables always exist, and if not, can one construct confidence intervals in some other way? Regarding the first question, the criteria for existence of pivotal random variables were given by Antle and Bain (1969). In analogy with examples concerning the normal distribution, we have the following theorem:

Theorem 11.7.2

- (a) If θ is a location parameter and $\hat{\theta}$ is MLE of θ , then $\hat{\theta} - \theta$ is pivotal.
- (b) If θ is a scale parameter, and $\hat{\theta}$ is MLE of θ , then $\hat{\theta}/\theta$ is pivotal.
- (c) Let θ_1 be a location parameter and θ_2 be a scale parameter so that $f(x, \theta_1, \theta_2) = (1/\theta_2)h[(x - \theta_1)/\theta_2]$ for some density h . If $\hat{\theta}_1$ and $\hat{\theta}_2$ are MLE's of θ_1 and θ_2 , respectively, then $(\hat{\theta}_1 - \theta_1)/\hat{\theta}_2$ is pivotal.

The answer to the second question is that even if pivotal random variables do not exist, we can construct confidence intervals. We will give here examples of confidence intervals for quantiles.

■ EXAMPLE 11.64

Let $X_1, \dots, X_n$ be a random sample from a distribution with cdf $F(t)$, and let $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$ be the order statistics of the sample. To simplify, assume that F is continuous and strictly increasing, and let $\xi_{1/2}$ be the median of the distribution F . Thus, $\xi_{1/2}$ is the unique solution of the equation $F(t) = 1/2$. We will construct confidence intervals for $\xi_{1/2}$. More precisely, we will consider (random) intervals of the form $[X_{a:n}, X_{b:n}]$ for integers a, b satisfying the condition $1 \leq a < b \leq n$ and assess, for each of such intervals, the probability that it covers the parameter $\xi_{1/2}$.

The event $X_{a:n} \leq \xi_{1/2}$ occurs if and only if at least a elements in the sample are below the median. Similarly, the event $\xi_{1/2} \leq X_{b:n}$ occurs if and only if at least $n - b$ elements in the sample are above the median. If we call an observation below the median a success and let S denote the number of successes in the sample, then the event $X_{n,a} \leq \xi_{1/2} \leq X_{n,b}$ occurs if and only if we have at least a successes and at least $n - b$ failures so that $a \leq S \leq b$. Since S has distribution $\text{BIN}(n, 1/2)$, we obtain

$$P\{X_{a:n} \leq \xi_{1/2} \leq X_{b:n}\} = \sum_{k=a}^b \binom{n}{k} \frac{1}{2^n}.$$

For instance, if $n = 10$, then the interval $[X_{3:10}, X_{7:10}]$ between the third and seventh order statistic is a confidence interval for the median with the confidence level

$$\sum_{k=3}^7 \binom{10}{k} \frac{1}{2^{10}} = 1 - 2 \left[\binom{10}{0} + \binom{10}{1} + \binom{10}{2} \right] \times \frac{1}{2^{10}} = 0.891.$$

On the other hand, the interval $[X_{2:10}, X_{6:10}]$ is a confidence interval for $\xi_{1/2}$ with confidence level $\sum_{k=2}^6 \binom{10}{k} (1/2)^{10} = 0.709$.

One-sided confidence bounds for $\xi_{1/2}$ are obtained in the same way: for instance, the order statistic $Y_{a:n}$ is a lower bound for $\xi_{1/2}$ with the confidence level $\sum_{k=a}^n \binom{n}{k} (1/2)^n$.

■ EXAMPLE 11.65

Continuing Example 11.64, let us now fix p and let ξ_p be the p th quantile of the population distribution. Again letting S be the number of successes, where now “success” is an observation below ξ_p , S has a $\text{BIN}(n, p)$ distribution. The interval $[X_{a:n}, X_{b:n}]$ is a confidence interval for ξ_p with the confidence level

$$\sum_{k=a}^b \binom{n}{k} p^k (1-p)^{n-k}.$$

Again, taking $n = 10$, the same interval $[X_{3:10}, X_{7:10}]$ can serve as a confidence interval for the first quartile $\xi_{1/4}$, except that now the confidence level will be

$$\sum_{k=3}^7 \binom{10}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{10-k} = 0.526.$$

However, in this case, it is better to take confidence intervals based on the order statistics where the indices are not equidistant from the extremes.

We complete this chapter with the critique of the concept of confidence intervals. The point is that while a pair of statistics $L = L(\mathbf{X}), U = U(\mathbf{X})$ may satisfy the condition

$$P\{U < \theta < L\} = 1 - \alpha,$$

there are cases where the sample $\mathbf{X}$ may provide additional information, allowing one to claim higher probability of coverage than $1 - \alpha$. This is illustrated by the following example from De Groot (1986).

■ EXAMPLE 11.66

Suppose that two independent observations X_1, X_2 are taken from the distribution uniform on $[\theta - 1/2, \theta + 1/2]$. Then order statistics $X_{1:2}$ and $X_{2:2}$ provide a 50% confidence interval for θ . Indeed, the interval $[X_{1:2}, X_{2:2}]$ covers θ if and only if one of

the observations is below θ and the other is above θ , an event with probability $1/2$. However, if $X_{2:2} - X_{1:2} > 1/2$, then the interval $[X_{1:2}, X_{2:2}]$ covers θ with probability 1. Still, strict adherence to the definition of confidence interval requires reporting such an interval as a 50% confidence interval rather than as a 100% confidence interval.

This example (as well as other examples of this type, e.g., Example 16.18) gives some weight to the Bayesian approach to statistical inference which will be discussed in Chapter 16.

Bootstrap Intervals

The bootstrap approach is a robust alternative to inference that is based on assumptions that may be either not valid or impossible to verify. For example, when sample sizes are too small, intervals based on asymptotic approximations can be inaccurate or calculations of standard errors can be very complex in some models. Bootstrap intervals can be obtained for the mean, median, proportion, correlation coefficient, regression coefficient, cdf—practically any population parameter. The quality of the procedure is not always the same, and therefore the effectiveness of the bootstrap approach has been extensively studied by carefully designed simulations for different parameters, types of population distributions, and sample sizes.

The idea underlying the bootstrap approach is that the sampling distribution of an estimator $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ can be “imitated” by the sampling distribution of the analogous estimator $\hat{\theta}^* = \hat{\theta}(X_1^{(B)}, \dots, X_n^{(B)})$ applied to so-called *bootstrap samples*.

This explains the name of the methodology which was introduced by Efron in his famous paper (Efron, 1979). Although “pulling yourself up by your bootstraps,” is in itself an impossible task, it is an idiomatic expression used to describe situations when difficult problems are solved by one’s own efforts, without any external help. Although such methods were proposed earlier²⁴ it was Efron who widened their applicability by utilizing modern computational power and who popularized the methodology so that it became an indispensable tool of modern data analysis.

Let us assume that $\mathbf{x} = (x_1, x_2, \dots, x_n)$ is the original sample consisting of n values randomly selected from $f(x, \theta)$ distribution, and let $\hat{\theta} = \hat{\theta}(\mathbf{x})$, be a value of an estimator selected to estimate unknown parameter θ .

The bootstrap random variable, $X^{(B)}$, is defined as

$$P\{X^{(B)} = x_i\} = \frac{1}{n}, \quad i = 1, \dots, n, \quad (11.105)$$

and each bootstrap sample, $\mathbf{x}_j^*$, $j = 1, \dots, B$, will be a random sample from the distribution (11.105) (in practice, this means that we are sampling the elements of the original sample with replacement). Next for each of the B bootstrap samples, we obtain the estimate

$$\hat{\theta}_j^* = \hat{\theta}(\mathbf{x}_j^*), \quad j = 1, \dots, B.$$

It has to be mentioned that the number B of bootstrap samples needs to be rather large, usually at least a thousand.

We introduce two main kinds of bootstrap interval estimation.

Bootstrap t Confidence Interval This interval is especially recommended when the distribution of $\hat{\theta}^*$ is close to normal, which can be verified by a Q - Q plot or any test for normality. The *bootstrap standard error* is estimated as

$$SE_{\text{BOOT}} = \sqrt{(1/B) \sum (\hat{\theta}_j^* - \hat{\theta}_{\text{BOOT}})^2},$$

²⁴References to earlier similar ideas can be found in Hall (1992).

where $\hat{\theta}_{\text{BOOT}} = \frac{1}{B} \sum_{j=1}^B \hat{\theta}_j^*$, and consequently, the $(1 - \alpha)100\%$ bootstrap t confidence interval has the form

$$\hat{\theta} \pm t_{\alpha/2, n-1} \text{SE}_{\text{BOOT}}.$$

Bootstrap Percentile Interval This interval does not require normality of bootstrap estimator $\hat{\theta}^*$. Its endpoints are sample quantiles of order $\alpha/2$ and $1 - \alpha/2$ obtained from $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$.

One of the advantages of the bootstrap percentile interval is that it is invariant to monotone transformations. If $(\theta_{1-\alpha/2}^*, \theta_{\alpha/2}^*)$ is a $(1 - \alpha)100\%$ interval for θ then

$$(h(\theta_{1-\alpha/2}^*), h(\theta_{\alpha/2}^*))$$

is a $(1 - \alpha)100\%$ interval for $\gamma = h(\theta)$ if h is monotone. No separate simulations need to be done.

The natural question now is: Is there any difference in these procedures, and if yes, then how to decide which one to use? We recommend that you obtain both, the bootstrap t confidence interval and the bootstrap percentile interval, and compare them. If they are not close, then probably neither of them should be used. If the bootstrap estimate of bias, $\hat{\theta} - \hat{\theta}_{\text{BOOT}}^*$, is small, then a bootstrap percentile interval is recommended. Also, since almost all variation in the bootstrap sample is caused by the selection of the original sample (resampling adds very little variation), bootstrap intervals based on small samples can be unreliable. For statistics such as the median or quartiles, they should be used with caution even for moderate samples. On the other hand, bootstrap t confidence intervals for the population mean θ , based on a trimmed mean as an estimator $\hat{\theta}$, can be a good choice.

In Example 11.67, we illustrate both types of bootstrap intervals.

■ EXAMPLE 11.67

We compare different confidence intervals for an unknown population mean θ in the $\text{EXP}(1/\theta)$ distribution. Initially, we assume a certain value of θ and use it to generate a random sample of size n (we select $n = 20$, and $\theta = 1$). A generated random sample will later be used to generate bootstrap samples. As an estimator $\hat{\theta}$ of a population mean θ we take the sample mean $\hat{\theta} = \bar{X}$.

The generated random sample $x_1, \dots, x_{20}$ yields a sample mean $\hat{\theta} = 1.1167$. Assuming that the information about the population distribution (exponential) is available, we apply formula (11.90) from Example 11.58 and obtain

$$\frac{\chi_{0.975, 40}^2}{2 \times 20 \times 1.1167} < \frac{1}{\theta} < \frac{\chi_{0.025, 40}^2}{2 \times 20 \times 1.1167},$$

or equivalently $0.7527 < \theta < 1.828$. After generating $B = 1,000$ bootstrap samples, we obtain $\hat{\theta}_{\text{BOOT}}^* = 1.1083$ and $\text{SE}_{\text{BOOT}} = 0.2038$. The distribution of the bootstrap estimator $\hat{\theta}^*$ does not indicate lack of normality. Consequently, we obtain 95% bootstrap t confidence interval as $1.1167 \pm 2.093 \times 0.238$, or $(0.6901, 1.5433)$, and the 95% bootstrap percentile interval as $(0.7478, 1.5551)$. All three intervals cover the true value $\theta = 1$, and their widths are 1.0753, 0.8532, and 0.8073, respectively. Bootstrap percentile interval seems to work very well here, but this should not be surprising, since, as mentioned, the bootstrap percentile interval should be recommended when the estimated bias is small. Here $\theta - \hat{\theta}_{\text{BOOT}}^* = 1.1167 - 1.1083 = 0.0084$ —less than 1% of a true value of the parameter.

Several methods to make bootstrap intervals more accurate have been proposed in the literature. We will not discuss them here, but we refer readers interested in the topic to Efron

and Tibshirani (1993) and Davison and Hinkley (1997). In Chapter 12, we will use bootstrap techniques for testing hypotheses.

PROBLEMS

- 11.7.1** Let $X_1, \dots, X_n$ be a random sample from the distribution with density $f(x; \theta) = \theta(x+1)^{-\theta-1}$ for $x > 0, \theta > 1$. Find the exact 90% confidence interval for θ .
- 11.7.2** Let $X_1, \dots, X_{16}$ and $Y_1, \dots, Y_{12}$ be random samples from $N(\theta, 15)$ and $N(\theta, 20)$ distributions, respectively. Find the 95% CI for θ if $\bar{x} = 20$ and $\bar{y} = 18$.
- 11.7.3** Let $X_1, \dots, X_n$ be a random sample selected from the Pareto distribution with density $f(x; \theta) = \theta 2^\theta x^{-(\theta+1)}$ for $x \geq 2$. Find: **(i)** The sufficient statistic for θ . **(ii)** An exact 95% CI for θ .
- 11.7.4** Based on a random sample 1.23, 0.36, 2.13, 0.91, 0.16, 0.12 selected from the $\text{GAM}(2.5, \theta)$ distribution, find an exact 95% CI for parameter θ .
- 11.7.5** Seven measurements of the concentration of some chemical in cans of tomato juice are 1.12, 1.18, 1.08, 1.13, 1.14, 1.10, 1.07. Assume that these numbers represent a random sample from the distribution $N(\theta, \sigma^2)$. **(i)** Find the shortest 95% and 99% CI's for θ , if σ^2 is unknown. **(ii)** Answer part (i) if $\sigma^2 = 0.0004$. **(iii)** Use the data to obtain a 90% CI for σ^2 and for σ .
- 11.7.6** A large company wants to estimate the fraction p of its employees who participate in a certain health program. It has been decided that if p is below 25%, a special promotion campaign will be launched. In a random sample of 85 employees the number of those who participated in the program was 16. **(i)** Find a 95% CI for p using formulas (11.94) and (11.95). Should the campaign be launched? **(ii)** Answer the same question if the data are 340 and 64, respectively.
- 11.7.7** Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ distribution with both parameters unknown. Let L_α be the length of the shortest confidence interval for μ on confidence level $1 - \alpha$. **(i)** Find $E(L_\alpha^2)$ as a function of n, σ^2 and α . **(ii)** Find the smallest n such that $E(L_\alpha^2) \leq \sigma^2/2$ for a given α .
- 11.7.8** Obtain a $(1 - \alpha)100\%$ CI for θ if $X_1, \dots, X_n$ is a random sample from a: **(i)** $N(\mu, \theta)$ distribution with μ and θ unknown. **(ii)** $\text{BETA}(1, \theta)$ distribution.
- 11.7.9** Find the probability that the length of a 95% confidence interval for the mean of normal distribution with unknown σ is less than $\sigma, n = 25$.
- 11.7.10** Suppose that the largest observation recorded in a sample of size $n = 35$ from a distribution uniform on $[0, \theta]$ is 5.17. Find a 90% CI for θ .
- 11.7.11** **(i)** Use the large sample distribution of MLE of mean λ in Poisson distribution to construct an approximate $(1 - \alpha)$ -level CI for λ . **(ii)** Assuming that the numbers of new cars of a given make sold per week in 15 consecutive weeks—5, 5, 6, 3, 5, 8, 1, 4, 7, 7, 5, 4, 3, 0, 9—form a random sample from a $\text{POI}(\lambda)$ distribution, find a 90% CI for λ .
- 11.7.12** Based on a random sample of size n selected from the $\text{WEI}(4, \theta)$ distribution, derive a 95% confidence interval for θ based on $X_{1:n}$.
- 11.7.13** Suppose that the lifetime T of a certain kind of device (e.g., a fuel pump in a car) has an $\text{EXP}(\lambda)$ distribution. The observed lifetimes of a sample of the devices are

350,727, 615,155, 962 (in days). Find a 95% CI's for: **(i)** λ . **(ii)** $E(T)$. **(iii)** The standard deviation of the lifetime of the device. **(iv)** The probability that the two copies of the device that you bought will each last more than two years.

- 11.7.14** Based on a random sample $X_1, \dots, X_n$ from the $U[0, \theta]$ distribution find: **(i)** $P\{X_{n:n} < \theta < 2X_{n:n}\}$. **(ii)** Such k that the interval $(X_{n:n}, kX_{n:n})$ is a $(1 - \alpha)100\%$ CI for θ .
- 11.7.15** Let $X_1, \dots, X_n$ be a random sample from some continuous distribution. If the confidence interval for the median M is $(X_{1:n}, X_{n:n})$ find the smallest n such that the confidence level $1 - \alpha$ is at least 0.95.
- 11.7.16** Let $X_1, \dots, X_n$ be a random sample from $\text{EXP}(\lambda)$ distribution. **(i)** Find an exact $100(1 - \alpha)\%$ confidence interval for $\theta = 1/\lambda$. **(ii)** Assuming that you have m such intervals, each based on a random sample of the same size n , find the probability that at least k of them ($k < m$) contain θ .
- 11.7.17** Suppose that the arrivals at a checkout counter in a supermarket (i.e., times of arriving at the counter or joining the queue, whichever is earlier) form a Poisson process with arrival rate λ . Counting from noon, the thirteenth customer arrived at 12:18 p.m. Find a 90% CI for: **(i)** λ . **(ii)** The variance of interarrival times between consecutive customers.
- 11.7.18** A sample of 200 trees in a forest has been inspected for a presence of some bugs, out of which 37 trees were found to be infested. **(i)** Assuming a binomial model, give a 90% confidence interval for the probability p of a tree being infested. Use the exact and the approximate formulas. **(ii)** Usually, if a tree is infested, one might expect some of the neighboring trees to be infested too. Thus, whether the binomial model is adequate depends on the way the sample was selected. Describe a way of selecting 200 trees so that the binomial model is realistic.
- 11.7.19** **(i)** Given a random sample $X_1, \dots, X_n$ obtain a 90% CI for the mean μ and then obtain a 90% bootstrap t interval and a 90% percentile interval for the same parameter μ . In addition, obtain two bootstrap intervals (t and percentile) for μ using a 10% trimmed mean as a statistic. As your sample use a data set of size $n = 20$ generated from the $N(1, 1)$ distribution. Use 2,000 bootstrap samples. Compare all intervals. **(ii)** Follow the same steps as in (i) with the only difference that the data will now be generated from the $\text{EXP}(2)$ distribution. **(iii)** Compare results obtained in (i) and (ii).
- 11.7.20** Use the data in Problem 12.10.5 to obtain 90% bootstrap intervals (t and percentile) for the mean width-to-length ratio of Shoshoni handicraft. Use 2,000 bootstrap samples. As statistic use $\bar{X}$ and the 10% trimmed mean. Compare your results.